MMT Project Writeup - Replica method applied to a neuronal system

Bruno F

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Contents

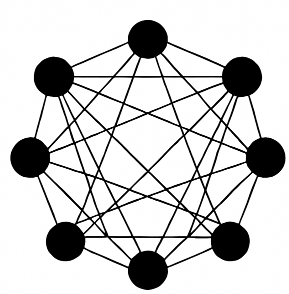
1 Introduction

Neural networks are the most important development in artificial intelligence in the last decade, as they provide a way to automate intelligence, turning tasks that required human perception and reasoning into computational problems. In fact, they are the engine of modern AI and their ability to learn from data is reshaping every industry on the planet.

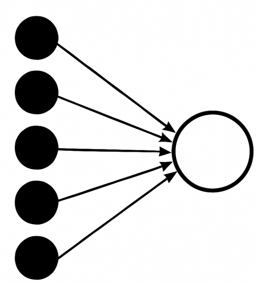
The objective of this study is to analyze two classical neural networks that model two fundamental processes: memory and decision-making. We will examine their behavior in terms of their capacity, that is, the ratio between the number of patterns that can be stored in the network and the number of neurons that compose it. From now on, this ratio will be denoted by α, and it will be the central focus of our project. But first, what is a neural network?

A neural network is a computing system consisting of interconnected processing elements (called "neurons" or "nodes") that process information by responding to external inputs and relaying information between each other. The network learns by adjusting these connections based on examples it is shown.

We can think of it as a human brain. Imagine how we recognize an object: at first we see examples of it; then we identify features of the object forming connections between our neurons strengthening pathways associated with it; eventually, we can recognize the object in other situations because our brain has learned essential patterns. We can represent them as follows:



Hopfield Network



Perceptron Network

It is very important to understand what a neuron is and how it works because neurons form the core of neural networks. A neuron is a simple computational unit inspired by biological brain cells. It receives one or more input values, multiplies them by associated weights, adds them together (often with a bias), and then passes the result through an activation function. The activation function determines the neuron's output, which is then sent to other neurons in the network. However, as there are thermal fluctuations affecting chemical release at synapses in biological neurons, something similar occurs with computational neurons: noise. Noise replaces the deterministic update rule of neurons with a stochastic one. In other words, neurons want to do what their inputs tell them, but there is also noise. A simple example of this behavior is the additive noise formulation:

$$\sigma_i = \operatorname{sgn}\left(\sum_{j\neq i} W_{ij}\,\sigma_j + \eta\right),$$

where σ_i is the state of the neuron i, W_{ij} represents the weights of the connections from the other neurons to the neuron i and η represents a random noise term. Notice that if the signal of the input ($\sum W\sigma$) is strong, the noise η is unlikely to be large enough to flip the sign, so the neuron behaves deterministically following the simple sign rule of the Hopfield dynamics:

$$\sigma_i = \operatorname{sgn}\left(\sum_{j\neq i} W_{ij}\,\sigma_j\right)$$

However, with a weak signal (near zero) the noise can easily dominate causing the neuron to flip state randomly.

In the Hopfield model, a slightly different noise model will be considered. The basic principle is the same (strong inputs are harder to flip), but with the formulation that will be used (see eg. page 10 in the experimental section), the equilibrium distribution will turn out to be the Boltzmann distribution, which is theoretically convenient and well studied.

As we already mentioned, we will focus on two different neural networks: the Hopfield Network and the Perceptron. We will explain how these networks behave and what they are used for. We will also explain and demonstrate how to obtain the critical α and, in particular, for certain values, as done in [3]. Finally, we will compare the results obtained theoretically with the ones that we will obtain by simulating these neural networks.

Nevertheless, in order to demonstrate this, we will use a technique from statistical physics, called the replica trick, used to compute disorder-averaged quantities that are otherwise difficult to evaluate, such as $\langle \ln Z\rangle$ for systems with quenched disorder. The method replaces the direct average of the logarithm with the identity

$$\ln Z = \lim_{n\to 0} \frac{Z^n - 1}{n},$$

allowing one to work instead with $\langle Z^n\rangle$, the partition function of n replicated copies of the system.

Applying the replica trick and modifying the resulting expression using standard mathematical manipulations, leads to a potential Φ that depends on several parameters such as α. The physical state of the neural network corresponds to the critical points (minima or saddle points) of this potential. Therefore, we will analyze the extrema of this potential Φ as a function of α to determine the behavior of the system.

2 Hopfield Network

2.1 Theory

A Hopfield network is a type of recurrent artificial neural network that functions as an associative memory system. Introduced by John Hopfield in 1982, it is designed to store patterns and retrieve them even when presented with partial or noisy inputs.

The network is fully connected, meaning that each neuron is linked to every other neuron except itself, allowing information to propagate across the entire system. Its neurons are binary, typically taking values in ± 1, and the synaptic weights are symmetric, so that the connection from neuron i to neuron j equals that from j to i. The dynamics can be understood through an energy-based formulation: the network evolves in time by monotonically decreasing an energy function and eventually settles into a stable state, or attractor.

To determine the critical value of α, we must therefore study this energy function, which in noiseless dynamics is guaranteed to decrease with every update. However, we are interested in the more general and physically relevant case where noise is present. In this stochastic setting, the probability of the network occupying a given state is governed by the Boltzmann distribution, making the effective energy landscape temperature-dependent. As we will see, this leads naturally to the minimization of the *free energy*, which incorporates both the energy of a state and the entropy (the number of microscopic configurations) associated with low-energy regions.

We will start with some definitions. We have N neurons connected through a weight matrix J. We will assume that the weight matrix J is symmetric ($J_{ij} = J_{ji}$).

The activity of the network evolves as follows: at each time step, a randomly selected neuron updates its activity to the sign of the total weighted activity of its inputs:

$$\sigma_i \to \operatorname{sgn}\left(\sum_{j \neq i} J_{ij}\sigma_j\right).$$

Note that we ignore the "self-input" $J_{ii}\sigma_i$ as a neuron is not connected to itself.

These dynamics can only decrease the total energy function

$$E := -\frac{1}{2}\sum_{i \neq j} J_{ij}\sigma_i\sigma_j. \tag{1}$$

The Hopfield rule for storing patterns $x^{(1)}, \ldots, x^{(P)}$ is to set the weight matrix to

$$J_{ij} = \frac{1}{N} \sum_{\mu=1}^{P} x_i^{(\mu)} x_j^{(\mu)}. \tag{2}$$

We will assume for ease of analysis that the patterns are independent and identically distributed (iid.), meaning that every entry of every pattern is ± 1 with probability $\frac{1}{2}$ each. We will write $\alpha := P/N$ for the ratio of patterns to neurons. This α determines whether the memory network will work. If α is too large, the network will not be able to recall the stored patterns.

Throughout this section, we will analyze a series of issues regarding the implementation of this model in order to understand how this neural network works and what capacity it has to store information.

As mentioned before, the deterministic Hopfield dynamics decrease the energy of the network state. When noise is introduced, transitions to higher-energy states become possible. As a result, the dynamics of the network set up a distribution over possible states of the network, the Boltzmann distribution, which puts more weight on low energy states and less on high energy states, so it is less probable that the network jumps to a higher energy state by mistake. To understand the typical behavior of the system, we study its free energy, which can be understood as the cumulant generating function of the Boltzmann distribution. This is

$$f := \mathbb{E}_{x^{(1)} \ldots x^{(p)}} \frac{1}{N} \log \sum_{\sigma \in \{-1,1\}^N} e^{-\beta H(\sigma)}, \tag{3}$$

where the energy function $H(\sigma)$ is defined as

$$H(\sigma) := -\frac{1}{2N} \sum_{\mu=1}^{P} (\sum_{i=1}^{N} x_i^{(\mu)} \sigma_i)^2. \tag{4}$$

This expression for $H(\sigma)$ is just the energy function (1) evaluated in the network state σ when the weights are set according to the Hopfield storage rule (2) (see 4.1.1).

The expression for the average free energy (3) contains a logarithm inside the expectation, which leads to difficulties in the computation. Therefore, we will apply a very common trick in statistical physics, the **replica trick**.

Lemma 1 (Replica Trick.) Let f be a positive random variable ($f > 0$ almost surely) defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Assume that there exists an open interval I containing 0 such that:

1. For all $n \in I$, the moment $\mathbb{E}[f^n]$ is finite.

2. The map $n \mapsto f^{\,n}$ is dominated by an integrable random variable in a neighborhood of $n = 0$, allowing differentiation under the expectation.

3. The function
$$\phi(n) := \log \mathbb{E}[f^{\,n}]$$
is differentiable at $n = 0$.

Then the following identity holds:

$$\mathbb{E}[\log f] \;=\; \lim_{n \to 0} \frac{1}{n} \log \mathbb{E}[f^{\,n}].$$

(Proof in 4.1.2)

Applying lemma 1 to our free energy (3) we obtain:

$$f = \lim_{n \to 0} \frac{1}{n} f^{(n)}$$

where
$$f^{(n)} = \frac{1}{N} \log \mathbb{E}_x \sum_{\sigma \in \{-1,1\}^{N \times n}} e^{\frac{\beta}{2N} \sum_{a=1}^{n} \sum_{\mu=1}^{P} (\sum_{i=1}^{N} x_i^{(\mu)} \sigma_i^a)^2} \tag{5}$$

The replica trick has allowed us to express the free energy in terms of n replicated systems. Following the replica methodology outlined at the end of the introduction (see 1), the free energy of the Hopfield network in the thermodynamic limit ($N \to \infty$) is determined by the extrema of the effective potential function Φ.

Theorem 1 (Saddle-Point equations for the Hopfield Model) Given a Hopfield network with N neurons, P i.i.d patterns $x^{(\mu)} \in \{-1, +1\}^N$, $\alpha = \frac{P}{N}$, applying the replica trick to its free energy yields the potential

$$\Phi^{(n)}(m, \hat{m}, q, \hat{q}) = \frac{1}{2}\hat{m}m - \frac{1}{4}\hat{q}q + \frac{\beta}{2}m^2 - \frac{\alpha}{2}\log(1 - \beta(1-q)) + \frac{\alpha}{2}\frac{\beta q}{1 - \beta(1-q)}, \tag{6}$$

whose critical points describe the behavior of the network.
The physical order parameters of the system (m and q) are found by solving the critical points of this potential. By imposing the stationary conditions ($\nabla\Phi = 0$ with respect to $m, \hat{m}, q, \hat{q}$), we obtain the saddle-point equations, which gives us the free energy of the system:

$$\begin{cases} m = \mathbb{E}_z\left[\tanh\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right] \\ q = \mathbb{E}_z\left[\tanh^2\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right], \end{cases} \tag{7}$$

where m is the overlap between the state of the network and some stored pattern x and q is the average overlap between replicas. In other words, if the network is started near some stored pattern x and the networks dynamics run for a long time until an approximate equilibrium, the m just measures the overlap with the pattern x.

See proof in 4.1.3.

Low temperature limit.
Assuming the memory functions correctly, there should be a strong minimum around the first stored pattern. It's reasonable to hypothesize that all replicas converge onto this same local minimum. This would mean perfect overlap of the different replicas as $\beta \to \infty$, that is, q $\to 1$.

Considering that Δ tends towards a constant, we will clear q and obtain $q = 1 - \frac{\Delta}{\beta}$. With some algebra we obtain the following:

$$1 - \beta(1-q) = 1 - \beta\left(1 - 1 + \frac{\Delta}{\beta}\right) = 1 - \Delta.$$

$$\frac{\alpha q}{1 - \beta(1-q)} = \frac{\alpha\left(1 - \frac{\Delta}{\beta}\right)}{1 - \Delta} \simeq \frac{\alpha}{1-\Delta} \quad \text{as } \beta \to \infty.$$

Taking into account (18) and (19), and modifying the second expression applying the β limit we obtain the following expression for Δ. We are now ready to analyse the $\beta \to \infty$ limit.

$$\boxed{\Delta \;=\; \beta\, \mathbb{E}_z\left[\operatorname{sech}^2\left(\beta\left(m + z\sqrt{\tfrac{\alpha q}{1-\beta(1-q)}}\right)\right)\right]}.$$

Using *Desmos* we can observe that $\tanh(\beta x) \to sg(x)$ as $\beta \to \infty$, so we can easily simplify (18).

$$\mathbb{E}_z\left[sg(m \;+\; z\frac{\sqrt{\alpha}}{1-\Delta}\right] = P_z(|z| \leq \frac{m(1-\Delta)}{\sqrt{\alpha}}),$$

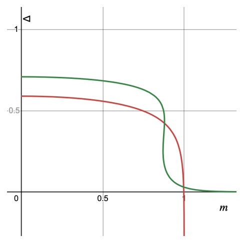
$$\boxed{m = P_z(|z| \leq \frac{m(1-\Delta)}{\sqrt{\alpha}})}.$$

Using *Desmos* again, we can now observe that $\beta \operatorname{sech}^2(\beta x) \to 2\delta(x)$ as $\beta \to \infty$ (details in 4.1.8). We simplify again the expected value expression.

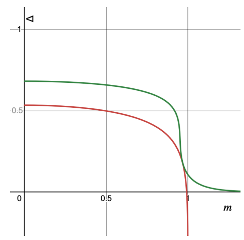
$$\mathbb{E}_z\left[2\,\delta(m \;+\; z\frac{\sqrt{\alpha}}{1-\Delta})\right] = \frac{2(1-\Delta)}{\sqrt{\alpha}}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}m^2\frac{(1-\Delta)^2}{\alpha}\right),$$

$$\boxed{\Delta = \frac{2(1-\Delta)}{\sqrt{\alpha}}\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}m^2\frac{(1-\Delta)^2}{\alpha}\right)}.$$

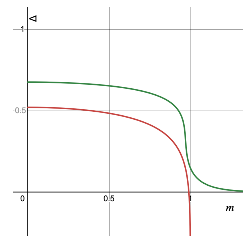
In the limit $\beta \to \infty$, corresponding to zero temperature, the dynamics of the network become fully deterministic. In this regime, when the storage load α is small, the fixed-point equations admit non-trivial solutions (with $m \approx 1$), meaning the system successfully retrieves the stored patterns. As α increases, the interference between patterns grows. This retrieval phase persists only up to a critical storage capacity $\alpha_c \approx 0.138$. Beyond this critical value, the non-trivial retrieval solutions vanish and the system collapses to a state with no macroscopic overlap with the memories ($m = 0$), indicating the complete loss of memory retrieval ability.



$0 < \alpha < \alpha^*$



$\alpha \approx \alpha*$



$\alpha > \alpha*$

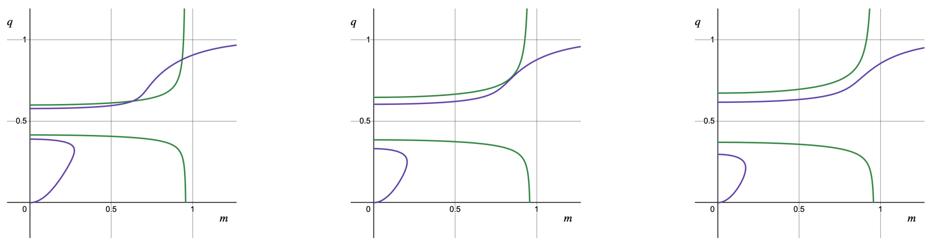
The following link provides an interactive Desmos visualization of the self-consistency equations in the zero-temperature case in the Hopfield model: https://www.desmos.com/calculator/bf46nswwyz?lang=es.

Finite-temperature case ($\beta = 2$).
We now move to the finite-temperature case with $\beta = 2$, where the dynamics are no longer deterministic. In this setting, thermal fluctuations introduce **noise** in the update rule, effectively reducing the stability of retrieval states. As a consequence, the critical value of α at which memory retrieval breaks down is expected to be lower than in the zero-temperature limit. We therefore study the same fixed-point equations under finite β to quantify how thermal noise affects the storage capacity of the network.

$$\begin{aligned} m &= \mathbb{E}_z\left[\tanh\Big(2\big(m + z\,\frac{\sqrt{\alpha q}}{1-2(1-q)}\big)\Big)\right], \\ q &= \mathbb{E}_z\left[\tanh^2\Big(2\big(m + z\,\frac{\sqrt{\alpha q}}{1-2(1-q)}\big)\Big)\right], \end{aligned} \tag{8}$$

To locate the critical value α^* at $\beta = 2$, we can use numerical procedures. For the moment, we will not carry out this computation here.
As a preliminary qualitative check, we also implemented the corresponding fixed-point equations in the *Desmos* graphical calculator. This allowed us to visualize the solution curves for different values of α at $\beta = 2$. From this graphical analysis, we observed that the retrieval solution ($m^* > 0$) disappears at approximately $\alpha^* \approx 0.06$, which is consistent with the expected reduction in storage capacity at

finite temperature. However, we will return to this point in the **Experimental Section**.



$0 < \alpha < \alpha^*$ $\qquad$ $\alpha \approx \alpha*$ $\qquad$ $\alpha > \alpha*$

An interactive Desmos visualization of the equations for the Hopfield model at finite β is provided at the following link: https://www.desmos.com/calculator/vv5fswfcd9?lang=es.

2.2 Experimental Part

This section presents the experimental validation of the theoretical results. The codes can be found in https://github.com/brunofernandez-blip/MMT/blob/6eda8bfff5f9c9ea3a2d1a64956944e070066785/Replica_method.ipynb. To achieve this, we implemented a fully connected neural network which will be used to see if the critical α that we have calculated in the previous section is the same as the one we obtain when running the neural network for the initial conditions $\beta = 2$ and $m_0 = 0.9$ with 1000 sweeps. We will also run the code for several values of N, the number of neurons, and compare the results obtained.

The methodology consists of the implementation of the asynchronous Glauber dynamics for our Hopfield model [3, 1].

First, given the values of N and α, we generate a matrix for the $P = \alpha * N$ random patterns $x^{(1)}, x^{(2)}, ..., x^{(P)}$, each of size N.

To test the network, we change a certain number of bits of the first pattern, which will modify the initial state of some neurons. Depending on the ratio of patterns to neurons, the input α, the network will "remember" the first pattern before it was changed. We consider that the network "remembers" the pattern successfully if the overlap $m = m(t)$ is greater than a certain threshold. The number of bits that are modified depends directly on the input m_0, as it is the overlap between the initial state and $x^{(1)}$.

Once we have done this, for every sweep we update all neurons in a random order. To update the i^{th} neuron σ_i, we calculate the input to the i^{th} neuron as follows:

$$h_i = W[i,:] \cdot \sigma$$

But this is computationally too expensive because W is a matrix of size PxN and we want to train large neural networks with many neurons. Note that we defined the overlap $m^{(\mu)}$ for the pattern $x^{(\mu)}$ as

$$m^{(\mu)} = \frac{1}{N} x^{(\mu)} \cdot \sigma$$

Using this, we can reduce the computation to a dot product of two vectors of size P:

$$h_i = X[i,:] \cdot m - \alpha \sigma_i$$

where X is the matrix generated by the P patterns.

Now, with the probability

$$p = \frac{1}{2}(1 + \tanh(\beta h_i))$$

which depends on β and h_i, we update the state of the neuron to 1 or -1.

However, in order to be able to use this formulation, we need to keep m updated as σ change. So, if the updated σ_i is equal to the previous one, we do not have to update m. Otherwise,

$$m_{t+1} = m_t + (\triangle\ \sigma)_i \frac{1}{N} X[i,:]$$

Note that the procedure of updating the neurons is asynchronous as the neurons are updated one at a time. Synchronous updates would be computationally faster (all neurons at the same time) as it uses vectorization products, which are faster in computational terms. However, asynchronous updates are necessary to correctly simulate Glauber dynamics.

Finally, we have run the Glauber Dynamics code for the β, m_0 and 1000 sweeps previously mentioned for a range values of α, from 0.01 to 0.08. And for each value of α we have repeated the procedure 10 times.

The following plots correspond to the overlap traces m for every sweep in every α and the equilibrium overlap for the different α values, each pair of plots depending on a certain value of N.

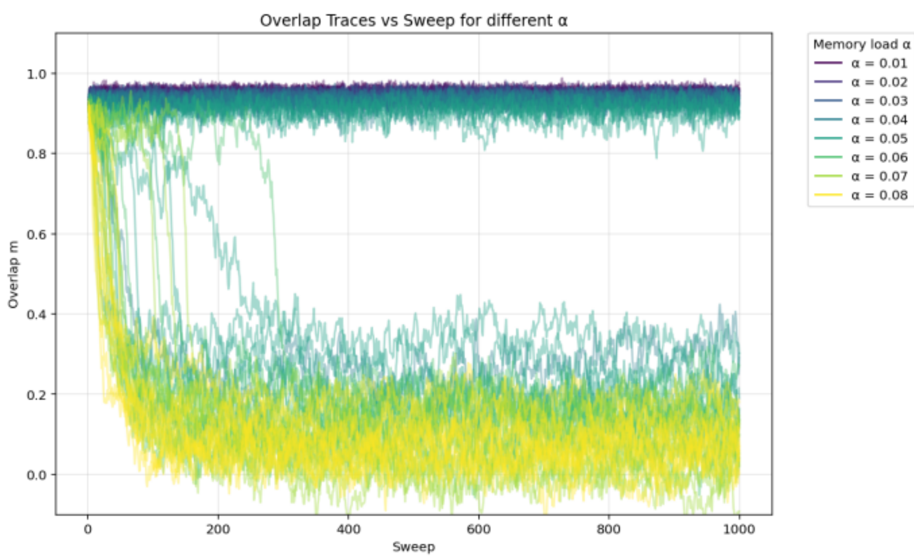


Figure 1: N=1000

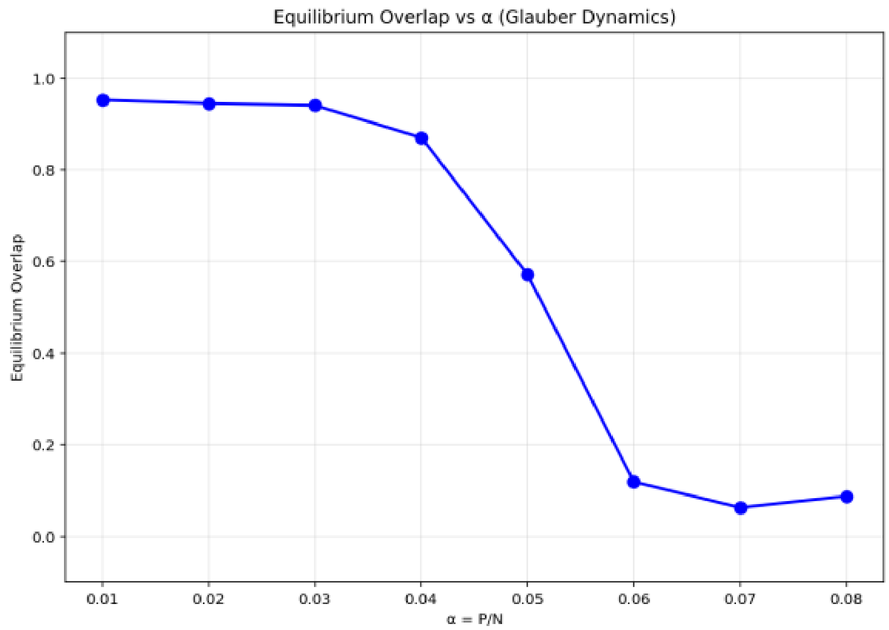


Figure 2: N=1000

As we can see in Figure 1, the overlap m remains in values around 0.9 when α is less than 0.06, which means that the neural network has "memory" and is able to restore the pattern $x^{(1)}$. However, for values of $\alpha \geq 0.06$, the overlap drops to almost 0 meaning that the network cannot remember the first pattern. This can be seen more clearly in Figure 2.

Note that the overlap does not suddenly decrease at $\alpha = 0.06$. Instead, it decreases gradually. The reason for this is "thermal noise" in Glauber dynamics (i.e. as the update of σ depends on a certain probability, sometimes the system flips spins against the direction that would reduce the energy. This creates small random changes, called fluctuations, that make the transition between high and low overlap smoother). A graphic demonstration of this can be seen in figure 1, when the overlap decreases in later sweeps (200,400...) and when the overlap is neither high enough nor small enough.

Figure 3: N=4000

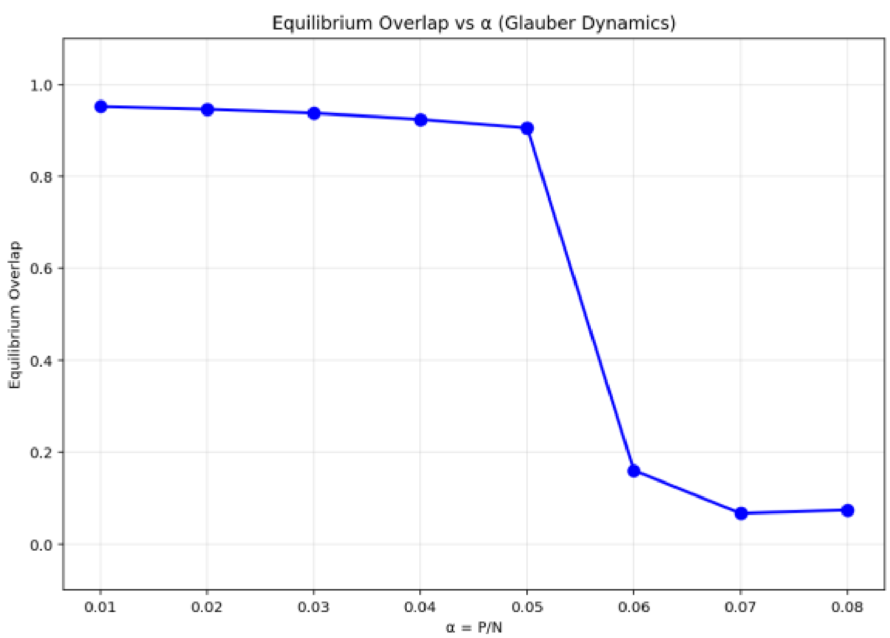


Figure 4: N=4000

In Figures 3 and 4, which correspond to $N = 4000$, the difference of the overlap between $\alpha = 0.05$ and $\alpha = 0.06$ is more notable.

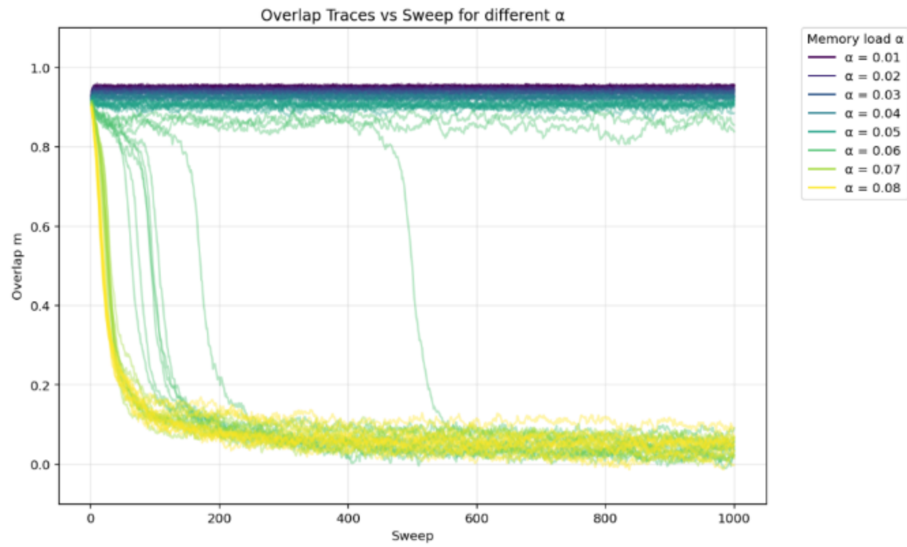


Figure 5: N=16000

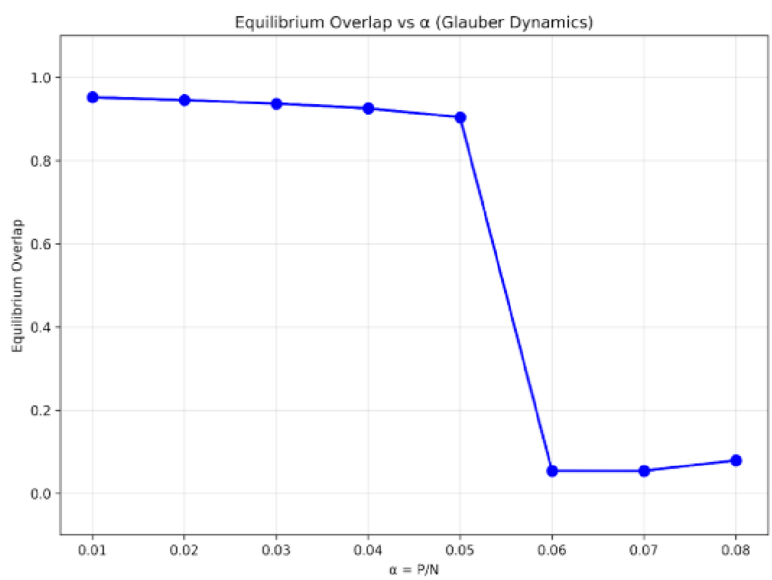


Figure 6: N=16000

As shown in Figures 5 and 6, the overlap drops to nearly zero at $\alpha = 0.06$ without any doubt. We therefore identify this value as the critical capacity, α_c, which aligns closely with theoretical predictions. Furthermore, the results demonstrate that as the number of neurons (N) increases, the precision of the network improves and the critical α value becomes more clearly defined.

Apart from this, another simulation has been run to observe the critical α when β is almost infinite, that is, when we do not have noise. Theoretical results approximate this value around $\alpha = 0.138$, which is similar to the one obtained in the following plot:

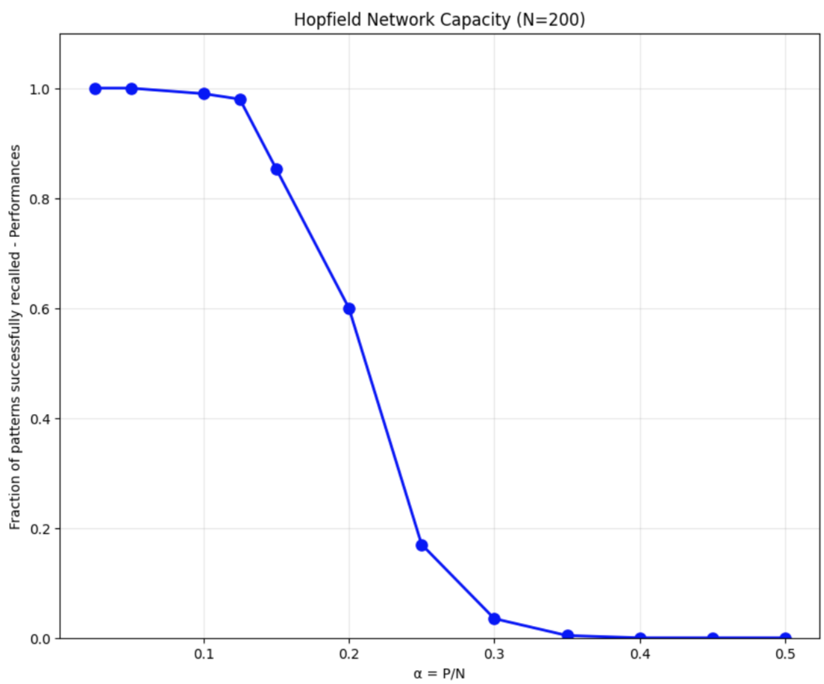


Figure 7: N=200

Also, we wanted to add some plots that show the complexity of the dynamics and demonstrates that synchronous update is faster than the asynchronous one. However, as mentioned before, the asynchronous update is necessary to correctly simulate the Glauber dynamics.

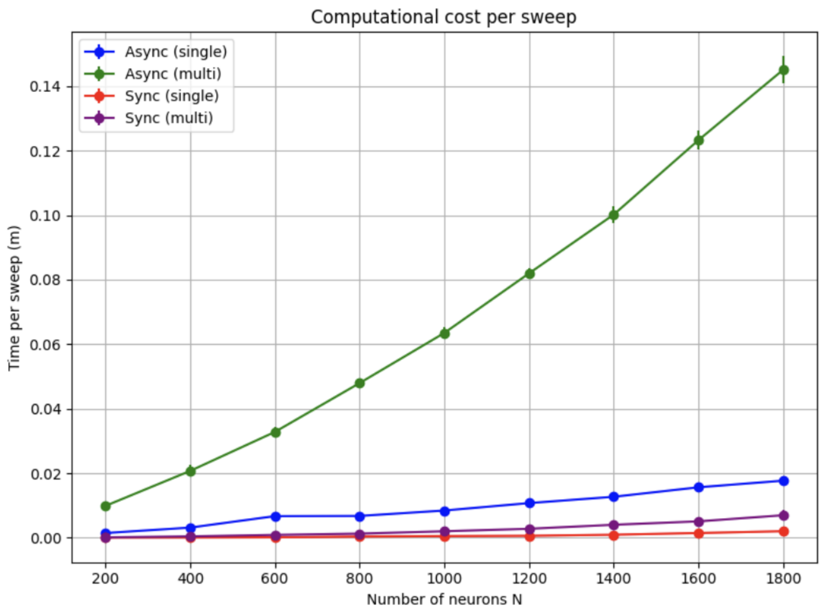


Figure 8: Total cost per sweep

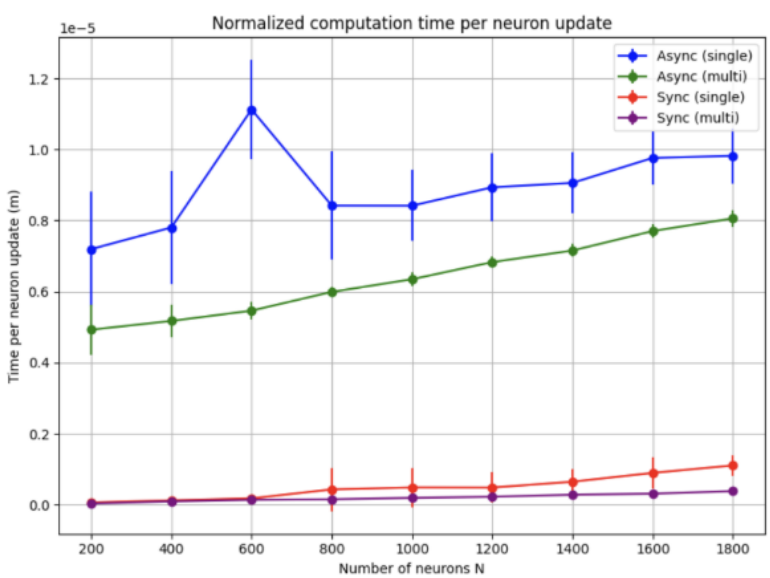


Figure 9: Total cost per neuron

3 Perceptron

3.1 Theory

A perceptron is the simplest form of an artificial neural network and is a basic building block for more complex models [2]. It is a supervised learning algorithm used for binary classification, meaning that it decides between two possible outputs.

A perceptron works by taking multiple input values, multiplying each by a corresponding weight, adding them together, and then passing the result through an activation function (usually a step function). If the weighted sum exceeds a certain threshold, the perceptron outputs one class (e.g., 1); otherwise, it outputs the other class (e.g., 0).

We are given inputs $x^{(1)}, ..., x^{(P)} \in \mathbb{R}^N$ and correct outputs $y^{(1)}, ..., y^{(P)} \in \{-1, 1\}$. We will assume $x^{(\mu)} \sim \mathcal{N}(0, I_N)$ for simplicity. Similarly to the Hopfield model, the perceptron comes with the activity rule $\hat{y} = \sigma(w \cdot x)$ (i.e. the output neuron adds up all inputs, weighted by the synaptic weights w, and fires $+1/-1$ if the total input is positive/negative.)

It is easy to show that the network correctly classifies the μ^{th} pattern iff

$$y^{(\mu)} \frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq 0.$$

We are interested in the slightly stronger condition that the network classifies its inputs correctly with confidence κ:

$$y^{(\mu)} \frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa.$$

Now we can state the fundamental question as follows:

Given N neurons and $P = \alpha N$ patterns, is there a setting of the weights w that correctly classifies all patterns with confidence κ?

Let us assume for convenience that $||w|| = \sqrt{N}$ (i.e. w is restricted to the sphere with radius $\sqrt{N}$, which we will write as S^{N-1}.)

We can answer our question fairly directly by simply integrating all volumes in S^{N-1} that correctly classifies all patterns:

$$V = \int_{S^{N-1}} dw \left[\prod_{\mu=1}^{P} \Theta \left(y^{(\mu)} \frac{w \cdot x^{(\mu)}}{\sqrt{N}} - \kappa \right) \right]$$

where Θ is the Heaviside step function. The Heaviside step function can be defined as follows:

$$\Theta(x) = \begin{cases} 0 & \text{si } x < 0 \\ 1 & \text{si } x \geq 0 \end{cases}.$$

We can slightly simplify this by flipping both $x^{(\mu)}$, $y^{(\mu)}$ whenever $y^{(\mu)} = -1$ (see justification in 4.2.1).

This gives:

$$V = \int_{S^{N-1}} dw \left[\prod_{\mu=1}^{P} \Theta \left(\frac{w \cdot x^{(\mu)}}{\sqrt{N}} - \kappa \right) \right].$$

Now we take logs, normalize and average over the patterns, obtaining:

$$\mathcal{V} := \mathbb{E}_x \frac{1}{N} log\ V = \mathbb{E}_x \frac{1}{N} log \int_{S^{N-1}} dw \left[\prod_{\mu=1}^{P} \Theta \left(\frac{w \cdot x^{(\mu)}}{\sqrt{N}} - \kappa \right) \right]. \tag{9}$$

Intuitively, $\mathcal{V}$ measures the volume (in logarithmic units) of the region in the w-space that correctly classifies all patterns. If $\alpha = \frac{P}{N}$ is too large, i.e, if we ask the network to correctly classify too many patterns, we expect the volume of the feasible set to tend toward 0, so $\mathcal{V}$, its logarithm, should tend toward $-\infty$.

Again, the logarithm inside the expectation leads to difficulties in the calculus of 9, so we can apply **Lemma 1 (Replica Trick)** and the expression becomes

$$\mathcal{V} = \lim_{n \to 0} \frac{1}{nN} \log \int \left[\prod_{a=1}^{n} dw^a \right] \mathbb{E}_x \left[\prod_{\mu=1}^{P} \prod_{a=1}^{n} \Theta \left(\frac{w^a \cdot x^{(\mu)}}{\sqrt{N}} - \kappa \right) \right]. \tag{10}$$

The replica identity states that $\mathbb{E}[\log V] = \lim_{n \to 0} \frac{1}{N} \log \mathbb{E}[V^n]$. We first calculate the replicated expression, adding n copies of the weight vector:

$$V^n = \int_{S^{N-1}} dw^1 \int_{S^{N-1}} dw^2 \cdots \int_{S^{N-1}} dw^n \prod_{\mu=1}^{P} \prod_{a=1}^{n} \Theta \left(\frac{w^a \cdot x^{(\mu)}}{\sqrt{N}} - \kappa \right).$$

Substituting and normalizing, we get what we expected.

□

What we have done will allow us to work with a more manageable expression when we later derive our results. We have been able to express the volume of the solution space in terms of n replicated systems. Following the replica methodology outlined at the end of the introduction (see 1), the normalized log-volume of the spherical perceptron in the thermodynamic limit ($N \to \infty$) is determined by the extrema of the effective potential function Φ.

Theorem 2. Perceptron Potential Formula
Consider a spherical perceptron with weight vector $w \in S^{N-1}$, input patterns $x^{(\mu)} \sim \mathcal{N}(0, I_N)$ and margin κ. Define the solution volume

$$V = \int_{S^{N-1}} dw \prod_{\mu=1}^{P} \Theta\left(\frac{w \cdot x^{(\mu)}}{\sqrt{N}} - \kappa\right),$$

and its disorder-averaged normalized logarithm

$$\mathcal{V} := \mathbb{E}_x \frac{1}{N} \log V.$$

In the limit $N \to \infty$, and under the replica-symmetric ansatz, $\mathcal{V}$ is determined by a potential depending only on the overlap order parameter $q \in [0, 1]$, given by

$$\Phi(q) = \alpha\, \mathbb{E}_z \left[\log \mathbb{P}_{u \sim \mathcal{N}(z\sqrt{q},\, 1-q)}(u \geq \kappa)\right] + \frac{1+q}{4(1-q)} + \frac{1}{2}\log(1-q), \tag{11}$$

where $z \sim \mathcal{N}(0, I_N)$. The critical points of Φ describe the behavior of the network. The physical order parameter of the system q is found by solving the critical points of this potential.

Proof: Same methodology as Theorem 1. Further details are given in 4.2.2

Our last expression 11 represents the normalized log volume of the solution space $\frac{1}{N}\mathbb{E}[\log V]$, and the order parameter q represents the overlap of two randomly selected solutions to the same problem. As α increases from 0 and the set of solutions shrinks, we expect two things to happen:

1. First, the volume of the solution set should tend towards 0, so the log volume $\frac{1}{N}\mathbb{E}[\log V] \to -\infty$. This is how we will know the capacity threshold has been reached.
2. Second, since the solution space is shrinking, it is reasonable to assume that all solutions are found in a small cluster, so the overlap of randomly selected solutions should tend toward 1, i.e. $q \to 1$.

Consistent with this, the potential 11, has three terms that are all well behaved, except at $q = 1$. Let us look at what we have. As $q \to 1^-$,

1. $\frac{1}{2}\frac{1}{1-q} \to \infty$. Since we are interested in $\frac{1}{N}\mathbb{E}[\log V] \to -\infty$, some other term must cancel this one.
2. $\frac{1}{2}\log(1-q) \to -\infty$, but only logarithmically. This term does not grow fast enough to cancel the pole in $\frac{1}{2}\frac{1}{1-q}$.

3. It is not immediately clear how the remaining term $\alpha\,\mathbb{E}_z \log \mathbb{P}\left[u \geq \frac{\kappa - \sqrt{q}\,z}{\sqrt{1-q}}\right]$ behaves as $q \to 1^-$, but one thing we can immediately say is that if it does have a pole, the residue is directly proportional to α. Thus, we might hope that (1) it has a negative pole, and (2) there is a critical α where this pole becomes strong enough to exactly cancel the positive pole of $\frac{1}{2}\frac{1}{1-q}$. It turns out that this is indeed the case, as we now show.

Let's study the asymptotics of $\log \mathbb{P}[u \geq x]$. As $q \to 1^-$, the right side of the inequality $\frac{k-\sqrt{q}z}{\sqrt{1-q}}$ can tend to ∞ or $-\infty$ (or 0 if $k = z$). So, our first step is to obtain asymptotics of the function $\psi(x) = \log \mathbb{P}[u \geq x]$.

We can easily argue that

$$\lim_{x \to -\infty} \psi(x) = 0,$$

since

$$\lim_{x \to -\infty} \mathbb{P}[u > x] = 1,$$

because we are integrating almost all the Gaussian mass. So,

$$\lim_{x \to -\infty} \psi(x) = \lim_{x \to -\infty} \log(1) = 0.$$

We can also easily show that

$$\psi(0) = \log \frac{1}{2}$$

since

$$\mathbb{P}[u > 0] = \frac{1}{2}$$

if we have a normal distribution.

Now, let us study

$$\lim_{x \to \infty} \psi(x).$$

We will first rewrite the $\psi(x)$ Gaussian expression to simplify future calculus:

$$\psi(x) = \log \int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}\, du = \log \left\{ \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}u^2\right) \frac{1}{x}\, dt \right\},$$

where

$$t = ux - x^2, \qquad du = \frac{1}{x} dt.$$

Expressing u^2 as $u^2 = u^2(x,t)$ we get:

$$u = x + \frac{t}{x} \quad \Rightarrow \quad u^2 = \left(x + \frac{t}{x}\right)^2 = x^2 + 2t + \left(\frac{t}{x}\right)^2.$$

Thus,

$$-\frac{1}{2}u^2 = -\frac{1}{2}x^2 - t - \frac{1}{2}\left(\frac{t}{x}\right)^2.$$

Substituting this into our expression:

$$\psi(x) = \log\left[\frac{1}{x}\, e^{-x^2/2} \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{t}{x}\right)^2\right)\, dt\right].$$

If we finally use log properties and

$$-t - \frac{1}{2}\left(\frac{t}{x}\right)^2 = \frac{1}{2}\left[\left(\frac{t}{x}\right)^2 + 2t\right],$$

we obtain:

$$\psi(x) = -\frac{1}{2}x^2 - \log x + \log \int_0^\infty \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{1}{2}\left(\left(\frac{t}{x}\right)^2 + 2t\right)\right] dt, \qquad x > 0.$$

We can justify in several ways that

$$\lim_{x\to\infty} \psi(x) = -\infty.$$

An approach is to analyze the asymptotic behavior of the expression obtained above: when x becomes very large, the exponential term $e^{-x^2/2}$ dominates and drives $\psi(x)$ to $-\infty$.

Another way is to apply directly the probabilistic definition. As in $x \to \infty$, we request that a standard normal random variable exceeds an extremely large value, which occurs with probability approximately equal to zero. Therefore,

$$\int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du \longrightarrow 0,$$

and consequently,

$$\psi(x) = \log\left(\int_x^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2}\, du\right) \longrightarrow \log(0) = -\infty.$$

Hence, the limiting value of the probability goes to 0, and thus $\psi(x)$ tends to $-\infty$ asymptotically.

The true purpose of this step is not to compute the limit of $\psi(x)$ itself, but rather to determine how fast $\psi(x)$ tends to $-\infty$ as $x \to \infty$. This asymptotic decay rate is essential because, in the full expression of the potential, there is another term that diverges as $q \to 1$, typically like $+\frac{1}{1-q}$.

To identify the critical value of α, we must compare these two divergences:

- the negative divergence coming from $\psi(x)$,

- the positive divergence coming from the $\frac{1}{1-q}$ term.

Only by determining the rate at which $\psi(x)$ goes to $-\infty$ can we predict the value of α for which these competing infinities cancel each other.

We consider the last expression and focus on the asymptotic behavior of the last term for large x. Inside the integral, the term $(t/x)^2$ becomes negligible when $x \to \infty$, so for any fixed t we have

$$\exp\left[-\frac{1}{2}\left(\left(\frac{t}{x}\right)^2 + 2t\right)\right] = e^{-t}\,\exp\left[-\frac{1}{2}\left(\frac{t}{x}\right)^2\right] \longrightarrow e^{-t}.$$

Moreover, the integrand is dominated by $\frac{1}{\sqrt{2\pi}}e^{-t}$, which is integrable on $[0,\infty)$. Therefore, by the dominated convergence theorem, the integral converges to

$$\int_0^\infty \frac{1}{\sqrt{2\pi}}e^{-t}\,dt = \frac{1}{\sqrt{2\pi}}.$$

Taking logarithms, we obtain the limiting contribution of the third term:

$$\log\left(\int_0^\infty \frac{1}{\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\left(\frac{t}{x}\right)^2 + 2t\right)\right]dt\right) \longrightarrow \log\left(\frac{1}{\sqrt{2\pi}}\right), \qquad x \to \infty.$$

Thus, the asymptotic form of $\psi(x)$ for large x is

$$\psi(x) = -\frac{1}{2}x^2 - \log x + \log\frac{1}{\sqrt{2\pi}} + o(1).$$

We now study the divergence of

$$\alpha\,\mathbb{E}_z \log\mathbb{P}\left(u \geq \frac{\kappa - \sqrt{q}\,z}{\sqrt{1-q}}\right) = \alpha\int Dz\,\psi(x(z,q))\,, \qquad x(z,q) = \frac{\kappa - \sqrt{q}\,z}{\sqrt{1-q}}.$$

We divide the integral according to the sign of $x(z,q)$. When $q \to 1^-$, the condition $x(z,q) < 0$ corresponds to $z > \kappa$. In this region $x(z,q) \to -\infty$ and therefore $\psi(x) \to 0$. Hence, this part of the integral stays finite and does not produce a pole.

The divergence comes from the region $z < \kappa$, where $x(z,q) \to +\infty$ as $q \to 1^-$. For large x we use the asymptotic form

$$\psi(x) = -\frac{1}{2}x^2 - \log x + \log\frac{1}{\sqrt{2\pi}} + o(1).$$

Substituting $x = x(z,q)$, the dominant term is

$$\psi(x(z,q)) = -\frac{1}{2(1-q)}\,(\kappa - z)^2 \;+\; \text{less singular terms}.$$

Therefore, the contribution from $z < \kappa$ behaves as

$$\alpha \int_{z<\kappa} Dz\, \psi(x(z,q)) = -\frac{\alpha}{2(1-q)} \int_{-\infty}^{\kappa} Dz\, (\kappa - z)^2 \;+\; O(\log(1-q)).$$

The potential contains another divergence,

$$\frac{1}{2(1-q)},$$

so the cancellation of poles at $q = 1$ requires

$$\frac{1}{2} = \frac{\alpha}{2} \int_{-\infty}^{\kappa} Dz\, (\kappa - z)^2.$$

We finally obtain the critical perceptron capacity:

$$\boxed{\alpha_c^{-1} = \int_{-\infty}^{\kappa} Dz\, (\kappa - z)^2}.$$

The following link provides an interactive Desmos visualization of the self-consistency equations arising in the perceptron model: https://www.desmos.com/calculator/dirydyn2dm?lang=es.

3.2 Experimental part

To validate the analytical results derived via the replica method in this Section, we performed numerical simulations of a continuous spherical perceptron (as $||w|| = \sqrt{N}$). The goal was to measure the storage capacity α_c as a function of the confidence parameter κ and to compare the algorithmic performance with the theoretical limit.

3.2.1 Methodology

We implemented a simulation in Python using the scikit-learn library, the code for which can be found in https://github.com/brunofernandez-blip/MMT/blob/6eda8bfff5f9c9ea3a2d1a64956944e070066785/Perceptron.ipynb. However, in this section, we are only going to explain briefly the methodology and present the important results. The simulation procedure followed the steps outlined below:

1. **Data Generation:**

   For a system size of $N = 500$ neurons, we generated $P = \alpha N$ input patterns $x^{(\mu)}$ drawn from a standard normal distribution $\mathcal{N}(0, 1)$. Binary labels $y^{(\mu)} \in \{-1, 1\}$ were assigned randomly.

2. **Training:**

   We attempted to find a weight vector w capable of classifying these patterns using Logistic Regression. We utilized the *LogisticRegression* solver with *fit-intercept=False* (to enforce a hyperplane through the origin) and a high inverse regularization parameter ($C = 100$) to approximate the hard-margin limit of a pure perceptron. Also, we imposed a large number of iterations to ensure that each simulation was successfully fitted.

3. **Validation and Scaling:**

   After training, the weight vector was extracted and normalized to unit length ($||w|| = 1$). We then verified the classification success by checking the stability condition for every pattern μ:

   $$y^{(\mu)} \frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$$

   We computed the "fractional accuracy" as the proportion of patterns satisfying this condition.

4. **Phase Space Exploration:**

   We repeated this process for a grid of parameters, varying the load $\alpha \in [0.02, 2.1]$ (in order to show the theoretical α_c when κ is 0, which is a well-known result) and the required confidence $\kappa \in [0.0, 0.3]$.

3.2.2 Results and Discussion

The results of the simulation reveal a sharp phase transition between a "satisfiable" phase (where a solution exists) and an "unsatisfiable" phase. Furthermore, the simulation results confirm the theoretical predictions regarding the storage capacity of the spherical perceptron. We present these findings in two key visualizations:

Figure 10: The Sharp Phase Transition (Line Plot) The first visualization plots the Fractional Accuracy against the load α for various fixed values of the confidence κ.

- **The $\kappa = 0$ Limit:** The blue curve, representing $\kappa = 0$, shows that the perceptron maintains 100% accuracy until the load reaches a critical value of $\alpha \approx 2.0$. At this point, the accuracy drops sharply to zero. This numerical result is in exact agreement with the theoretical prediction derived in the replica calculation: $\alpha_c(\kappa = 0) = 2$.
- **Effect of Stability:** As the required margin κ increases (colored lines moving from blue to cyan), the critical capacity decreases, indicating that enforcing higher confidence reduces the number of patterns the network can store.

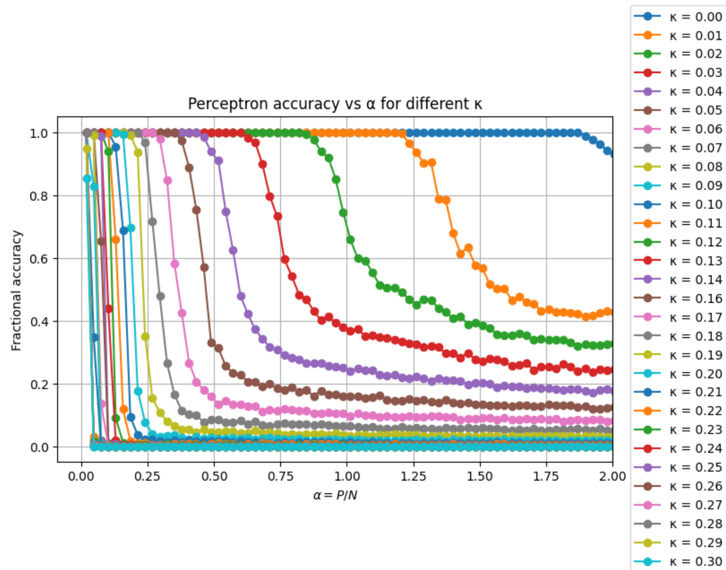


Figure 10: Perceptron accuracy vs α for different κ.

Figure 11: The Phase Diagram and Theoretical Bound (Heatmap) The second visualization provides a comprehensive Phase Diagram of the system.

- **Heatmap:** The background color represents the simulation accuracy (Yellow = 100% Solvable, Purple = 0% Solvable). The boundary between these regions represents the algorithmic capacity of our simulation.
- **Theoretical Overlay:** The red line represents the theoretical Gardner capacity curve, calculated by numerically solving the integral equation:

$$\alpha_c^{-1} = \int_{-\infty}^{\kappa} Dz\,(\kappa - z)^2$$

- **Agreement:** As seen in the figure, the red theoretical line perfectly hugs the boundary of the yellow "solvable" region. This confirms that our simulation (using Logistic Regression) is successfully finding solutions up to the exact theoretical limit predicted by Gardner's replica analysis.

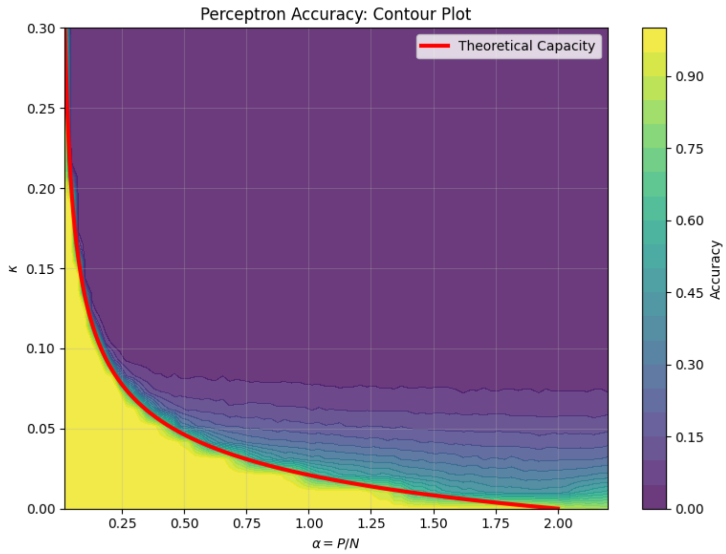


Figure 11: Phase Diagram and Theoretical Bound

The perfect overlap between the experimental boundary (11) and the theoretical curve confirms two key findings:

1. **Validity of the Replica Symmetric Ansatz:** For the spherical perceptron, the replica symmetric calculation yields the exact capacity.
2. **Algorithmic Success:** Unlike binary perceptrons, the continuous spherical perceptron has a convex solution space. Consequently, the standard Logistic Regression algorithm was able to find a solution whenever one theoretically existed, right up to the Gardner limit.

4 Appendix

4.1 Hopfield Network

4.1.1 Energy function

We take the total energy function (1) and substitute J_{ij} by it's expression (2):

$$E = -\frac{1}{2}\sum_{i\neq j} J_{ij}\sigma_i\sigma_j = -\frac{1}{2}\sum_{i\neq j}(\frac{1}{N}\sum_{\mu=1}^{P} x_i^{(\mu)}x_j^{(\mu)})\sigma_i\sigma_j = -\frac{1}{2N}\sum_{\mu=1}^{P}\sum_{i\neq j} x_i^{(\mu)}x_j^{(\mu)}\sigma_i\sigma_j$$

Let us now look at the equation $H(\sigma)$. If we expand the square and reunite this new expression with the one developed below, we obtain the following:

$$(\sum_{i=1}^{N} x_i^{(\mu)}\sigma_i)^2 = \sum_{i=1}^{N}\sum_{j=1}^{N}(x_i^{(\mu)}x_j^{(\mu)})\sigma_i\sigma_j = \sum_{i\neq j}(x_i^{(\mu)}x_j^{(\mu)})\sigma_i\sigma_j \ + \sum_{i=1}^{N}(x_i^{(\mu)}\sigma_i)^2,$$

where

$$x_i^{(\mu)}\sigma_i = \pm 1 \Rightarrow (x_i^{(\mu)}\sigma_i)^2 = 1 \Rightarrow \sum_{i=1}^{N}(x_i^{(\mu)}\sigma_i)^2 = \sum_{i=1}^{N} 1 = N.$$

Therefore,

$$H(\sigma) = -\frac{1}{2N}\sum_{\mu=1}^{P}(\sum_{i\neq j} x_i^{(\mu)}x_j^{(\mu)}\sigma_i\sigma_j + N) = -\frac{1}{2N}\sum_{\mu=1}^{P}\sum_{i\neq j} x_i^{(\mu)}x_j^{(\mu)}\sigma_i\sigma_j - \frac{P}{2}.$$

We can then conclude that $H(\sigma) = E + P/2$. We can discard the constant term $P/2$ because, as usual in dynamics, the evolution of the system depends on which state has lower energy relative to another, so we really do not care about additive constants because only the differences in energy between states matter, not the absolute value of the energy itself.

For example, the evolution towards an energy minimum does not rely on its exact value but rather on the comparison between different states.

4.1.2 Replica Trick Proof

Since $f > 0$ almost surely, the function $n \mapsto f^n$ is well defined. For n in a neighborhood of 0 we can write

$$f^n = e^{n \log f}.$$

Assumption (2) guarantees that we may differentiate inside the expectation, and assumption (1) guarantees that $\mathbb{E}[f^n]$ is finite.

Define

$$\phi(n) = \log \mathbb{E}[f^n].$$

Because $f^0 = 1$, we have
$$\phi(0) = \log \mathbb{E}[1] = 0.$$

By the chain rule and the ability to differentiate under the integral,
$$\phi'(n) = \frac{\mathbb{E}[f^{\,n} \log f]}{\mathbb{E}[f^{\,n}]}.$$

Evaluating at $n = 0$ gives
$$\phi'(0) = \frac{\mathbb{E}[1 \cdot \log f]}{\mathbb{E}[1]} = \mathbb{E}[\log f].$$

Now use the definition of the derivative of ϕ at 0:
$$\phi'(0) = \lim_{n\to 0} \frac{\phi(n) - \phi(0)}{n} = \lim_{n\to 0} \frac{1}{n} \log \mathbb{E}[f^n].$$

Combining the expressions for $\phi'(0)$ yields the identity
$$\mathbb{E}[\log f] = \lim_{n\to 0} \frac{1}{n} \log \mathbb{E}[f^{\,n}].$$

This proves the replica identity under the stated regularity conditions.

□

4.1.3 Theorem 1 proof

We start from the expression
$$f = \lim_{n\to 0} \frac{1}{n} f^{(n)}$$
where
$$f^{(n)} = \frac{1}{N} \log \mathbb{E}_x \sum_{\sigma \in \{-1,1\}^{N\times n}} e^{\frac{\beta}{2N} \sum_{a=1}^{n} \sum_{\mu=1}^{P} (\sum_{i=1}^{N} x_i^{(\mu)} \sigma_i^a)^2}.$$

We first simplify it by doing an average over the disorder (the random patterns).

Our strategy for studying the recall of a random pattern will be to examine the stability of the network's dynamics around a randomly selected pattern, say $x^{(1)}$. This stability will depend on the behavior of the energy function in the neighborhood of this pattern. So we will treat $x^{(1)}$ separately, only averaging over the remaining $P - 1$ patterns.

From the previous expression, we separate the contribution of the first pattern:

$$e^{\frac{\beta}{2N} \sum_{a=1}^{n} \sum_{\mu=1}^{P} (\sum_{i=1}^{N} x_i^{(\mu)} \sigma_i^a)^2} = e^{\frac{\beta}{2N} \sum_{a=1}^{n} (\sum_{i=1}^{N} x_i^{(1)} \sigma_i^a)^2} . e^{\frac{\beta}{2N} \sum_{a=1}^{n} \sum_{\mu=2}^{P} (\sum_{i=1}^{N} x_i^{(\mu)} \sigma_i^a)^2}.$$

Using that the patterns are i.i.d., the expectation factorizes as follows:

$$f^{(n)} = \frac{1}{N}\log\sum_{\sigma}\mathbb{E}_{x^{(1)}}\left[e^{\frac{\beta}{2N}\sum_{a=1}^{n}(\sum_{i=1}^{N}x_i^{(1)}\sigma_i^a)^2}\right]\cdot\mathbb{E}_{x^{(2)}\dots x^{(P)}}\left[e^{\frac{\beta}{2N}\sum_{a=1}^{n}\sum_{\mu=2}^{P}(\sum_{i=1}^{N}x_i^{(\mu)}\sigma_i^a)^2}\right] =$$

$$= \frac{1}{N}\log\sum_{\sigma}\mathbb{E}_{x^{(1)}}\left[e^{\frac{\beta}{2N}\sum_{a=1}^{n}(\sum_{i=1}^{N}x_i^{(1)}\sigma_i^a)^2}\right]\cdot\left(\mathbb{E}_{x}\left[e^{(\frac{\beta}{2N}\sum_{a=1}^{n}(x\cdot\sigma^a)^2)}\right]\right)^{P-1}. \quad (12)$$

We have considered the random variable x, which has the same distribution as the patterns.

We now have an expression that depends on the network states through their overlap with the patterns. The summand now depends on the spins σ only through a small number of quantities: the overlap with the first pattern $x^{(1)}\cdot\sigma^a$, and the overlap with the random patterns $x\cdot\sigma^a$ in the second factor. Since we are particularly interested in the overlap of network state σ with the first stored pattern, we introduce order parameters to characterize these overlaps:

$$m_a := \frac{1}{N}x^{(1)}\cdot\sigma^a, \quad a = 1,\dots,n.$$

This tracks how closely the a^{th} replica approximates the first stored memory.

Regarding the remaining overlap $x\cdot\sigma^a$, this appears in a term that is averaged over $x\in\mathbb{R}^N$, which has i.i.d entries with values ±1. By the central limit theorem, when N is large, the $u_a := \frac{1}{\sqrt{N}}x\cdot\sigma^a$ for $a = 1,\dots,n$ will be approximately jointly normal with mean and covariance

$$\mu_a := \mathbb{E}[u_a] = \frac{1}{\sqrt{N}}\sum_{i=1}^{N}\mathbb{E}[x_i]\sigma_i^a \quad \Rightarrow \quad \mathbb{E}[\sigma_i^a] = 0$$

$$Q_{ab} := \mathrm{Cov}(u_a,u_b) = \mathbb{E}[u_a u_b] - \mathbb{E}[u_a]\mathbb{E}[u_b]$$

$$\mathbb{E}[u_a u_b] = \frac{1}{N}\sum_{i,j}\mathbb{E}[x_i x_j]\sigma_i^a\cdot\sigma_j^b$$

$$\mathbb{E}[x_i x_j] = \begin{cases}1 & i=j\\ 0 & i\neq j\end{cases}$$

$$\Rightarrow \mathbb{E}[u_a u_b] = \frac{1}{N}\sum_{i=1}^{N}\sigma_i^a\sigma_i^b \quad \Rightarrow \quad Q_{ab} = \frac{1}{N}\sigma^a\cdot\sigma^b.$$

Using the definitions of m_a, u_a, Q, we can manipulate 12 as follows:

$$f^{(n)} = \frac{1}{N}\log\sum_{\sigma\in\{-1,1\}^{N\times n}}\mathbb{E}_{x^{(1)}}\left[e^{\beta\frac{1}{2N}\sum_{a=1}^{n}(\sum_{i=1}^{N}x_i^{(1)}\cdot\sigma_i^a)^2}\right]\left(\mathbb{E}_x\left[e^{\beta\frac{1}{2N}\sum_{a=1}^{n}(x\cdot\sigma^a)^2}\right]\right)^{P-1} =$$

$$= \frac{1}{N} \log \sum_{\sigma \in \{-1,1\}^{N \times n}} \left(\mathbb{E}_{x^{(1)}} e^{\beta \frac{N}{2} \sum_{a=1}^{n} m_a^2} \right) \left(\mathbb{E}_{u \sim \mathcal{N}(0,Q)} e^{\frac{\beta}{2N} \sum_{a=1}^{n} (\sqrt{N} u_a)^2} \right)^{P-1} =$$

$$= \frac{1}{N} \log \sum_{\sigma \in \{-1,1\}^{N \times n}} \mathbb{E}_{x^{(1)}} e^{\beta \frac{N}{2} \sum_{a=1}^{n} m_a^2} \left(\mathbb{E}_{u \sim \mathcal{N}(0,Q)} e^{\frac{\beta}{2} u^T u} \right)^{P-1},$$

giving

$$f^{(n)} = \frac{1}{N} \log \sum_{\sigma \in \{-1,1\}^{N \times n}} \mathbb{E}_{x^{(1)}} e^{\beta \frac{N}{2} \sum_{a=1}^{n} m_a^2} \left(\mathbb{E}_{u \sim \mathcal{N}(0,Q)} e^{\frac{\beta}{2} u^T u} \right)^{P-1}. \tag{13}$$

We make one last simplification to the gaussian expectation $\mathbb{E}_{u \sim \mathcal{N}(0,Q)} e^{\beta \frac{1}{2} u^\top u}$. The probability distribution function (pdf) of u is

$$p(u) = \frac{1}{\sqrt{|2\pi Q|}} e^{-\frac{1}{2} u^\top Q^{-1} u},$$

where $|\cdot|$ denotes the determinant, and so the expectation is simply

$$\mathbb{E}_{u \sim \mathcal{N}(0,Q)} e^{\beta \frac{1}{2} u^\top u} = \int p(u) e^{\beta \frac{1}{2} u^\top u} du = \frac{1}{\sqrt{|2\pi Q|}} \int e^{-\frac{1}{2} u^\top Q^{-1} u} e^{\beta \frac{1}{2} u^\top u} du.$$

If we calculate the value of the integral, we get the expression (see details in 4.1.4)

$$f^{(n)} = \frac{1}{N} \log \mathbb{E}_{x^{(1)}} \sum_{\sigma \in \{-1,1\}^{N \times n}} e^{\beta \frac{N}{2} \sum_{a=1}^{N} m_a^2 - \frac{P-1}{2} \log(|I - \beta Q|)}. \tag{14}$$

Up to this point, the introduction of the overlap parameters

$$m_a := \frac{1}{N} x^{(1)} \cdot \sigma^a, \quad 1 \leq a \leq n$$

$$Q_{ab} := \frac{1}{N} \sigma^a \cdot \sigma^b, \quad 1 \leq a < b \leq n,$$

is merely cosmetic; after all, the m's and Q's are functions of the $\sigma^1, \ldots, \sigma^n$ and substituting their definitions back into the energy would put us right back where we were before. (A justification of the restrictions imposed on the parameters a and b can be found in 4.1.5). So we will proceed as follows.

Given an energy function such as

$$F(\frac{1}{N}x^{(1)}\sigma^a) = \int \delta(m_a - \frac{1}{N}x^{(1)}\sigma^a)\, F(m_a)\, dm_a,$$

we integrate this expression, over all m_a, with a proper δ function, which will be its Fourier representation:

$$F(\frac{1}{N}x^{(1)}\sigma^a) = \int\int \frac{dm_a d\hat{m}_a}{2\pi/N} e^{\frac{N}{2}\hat{m}_a(m_a - \frac{1}{N}x^{(1)}\sigma^a)} F(m_a).$$

We want to apply this trick to decouple functions m and Q from the summation of our energy function. Therefore, our δ functions for m's and Q's will be, respectively,

$$\delta(m_a - \frac{1}{N}x^{(1)}\sigma^a) = \frac{N}{2\pi}\int d\hat{m}_a\, e^{\frac{N}{2}\hat{m}_a(m_a - \frac{1}{N}x^{(1)}\sigma^a)},$$

$$\delta(Q_{ab} - \frac{1}{N}\sigma^a\sigma^b) = \frac{N}{2\pi}\int d\hat{Q}_{ab}\, e^{\frac{N}{2}\hat{Q}_{ab}(Q_{ab} - \frac{1}{N}\sigma^a\sigma^b)}.$$

Finally, we substitute each of these values $m_a, \hat{m}_a, Q_{ab}, \hat{Q}_{ab}$ in our energy function 14:

$$f^{(n)} = \frac{1}{N}\log \mathbb{E}_{x^{(1)}} \sum_{\sigma\in\{-1,1\}^{N\times n}} e^{\left(\frac{N\beta}{2}\sum_{a=1}^{n} m_a^2 - \frac{P-1}{2}\log|I-\beta Q|\right)} =$$

$$= \frac{1}{N}\log \mathbb{E}_{x^{(1)}} \sum_{\sigma\in\{-1,1\}^{N\times n}} \int \left[\prod_{a=1}^{n} \frac{dm_a d\hat{m}_a}{2\pi/N}\right]\left[\prod_{1\le a<b\le n} \frac{dQ_{ab}d\hat{Q}_{ab}}{2\pi/N}\right].$$

$$\cdot\, e^{\left(\frac{N}{2}\sum_{a=1}^{n}\hat{m}_a\left(m_a - \frac{1}{N}x^{(1)}\cdot\sigma^a\right) + \frac{N}{2}\sum_{a<b}\hat{Q}_{ab}\left(Q_{ab} - \frac{1}{N}\sigma^a\cdot\sigma^b\right) + \frac{N\beta}{2}\sum_{a=1}^{n} m_a^2\ - \frac{P-1}{2}\log|I-\beta Q|\right)}. \tag{15}$$

With this expression, we now have the m's and the Q's decoupled from the summation.

This last expression involves a sum over the nN variables σ_i^a, $i = 1, \ldots, N$ and $a = 1, \ldots, n$, which each range over the values $\{-1, 1\}$; inside this is an integral over the $2n$ variables $m_a, \hat{m}_a$ and the $n^2 - n$ variables $Q_{ab}, \hat{Q}_{ab}$. Even though the exponent in the integrals/sums grows with N, we still can not apply the saddle point approximation because the dimensionality of the sum also grows with N.

In 15, not all the terms in the exponent depend on σ, so we can write the expression as

$$f^{(n)} = \frac{1}{N}\log \mathbb{E}_{x^{(1)}} \sum_{\sigma\in\{-1,1\}^{N\times n}} \int \left[\prod_{a=1}^{n} \frac{dm_a d\hat{m}_a}{2\pi/N}\right]\left[\prod_{1\le a<b\le n} \frac{dQ_{ab}d\hat{Q}_{ab}}{2\pi/N}\right]$$
$$e^{\frac{N}{2}\sum_{a=1}^{n}\hat{m}_a m_a + \frac{N}{2}\sum_{a<b}\hat{Q}_{ab}Q_{ab} + N\frac{\beta}{2}\sum_{a=1}^{n} m_a^2 - \frac{P-1}{2}log|I-\beta Q|}$$
$$\underbrace{e^{-\frac{1}{2}\sum_{i=1}^{N}(\sum_{a=1}^{n}\hat{m}_a x_i^{(1)}\sigma_i^a + \sum_{a<b}\hat{Q}_{ab}\sigma_i^a\sigma_i^b)}}_{(*)}$$

Only the (*) part depends on σ, so we can bring the sum over σ inside. Similarly, we can move the expectation over $x^{(1)}$ inside as well, yielding

$$f^{(n)} = \frac{1}{N}\log \int \left[\prod_{a=1}^{n} \frac{dm_a d\hat{m}_a}{2\pi/N}\right]\left[\prod_{1\le a<b\le n} \frac{dQ_{ab}d\hat{Q}_{ab}}{2\pi/N}\right]\cdot$$
$$\cdot e^{\frac{N}{2}\sum_{a=1}^{n}\hat{m}_a m_a + \frac{N}{2}\sum_{a<b}\hat{Q}_{ab}Q_{ab} + N\frac{\beta}{2}\sum_{a=1}^{n} m_a^2 - \frac{P-1}{2}\log|I-\beta Q|}\cdot$$
$$\cdot \mathbb{E}_{x^{(1)}} \sum_{\sigma\in\{-1,1\}^{N\times n}} \prod_{i=1}^{N} e^{-\frac{1}{2}\sum_{a=1}^{n}\hat{m}_a x_i^{(1)}\sigma_i^a - \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}\sigma_i^a\sigma_i^b}.$$

The last sum now factorizes over the site index i. We now have the expectation over $x^{(1)}$ of a product over i. As $x^{(1)}$ has independent entries, the expectation of the product is the product of the expectations of each entry. Taking into account that the entries are identically distributed (as a variable x that takes values in $\{-1,+1\}$ with probability $\frac{1}{2}$), all expectations will be equal. Therefore,

$$\mathbb{E}_{x^{(1)}} \prod_{i=1}^{N} \sum_{\sigma\in\{-1,1\}^{N\times n}} e^{-\frac{1}{2}\sum_{a=1}^{n}\hat{m}_a x_i^{(1)}\sigma_i^a - \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}\sigma_i^a\sigma_i^b} =$$
$$(\mathbb{E}_x \sum_{\sigma\in\{-1,1\}^{N\times n}} e^{-\frac{1}{2}\sum_{a=1}^{n}\hat{m}_a x\sigma^a - \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}\sigma^a\sigma^b})^N =$$
$$e^{N\log\left(\mathbb{E}_x \sum_{\sigma\in\{-1,1\}^{N\times n}} e^{-\frac{1}{2}\sum_{a=1}^{n}\hat{m}_a x\sigma^a - \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}\sigma^a\sigma^b}\right)}.$$

Plugging it into the previous expression, we see that 15 can be written as

$$f^{(n)} = \frac{1}{N}\log \int \left[\prod_{a=1}^{n} \frac{dm_a d\hat{m}_a}{2\pi/N}\right]\left[\prod_{1\le a<b\le n} \frac{dQ_{ab}d\hat{Q}_{ab}}{2\pi/N}\right]\cdot$$
$$\cdot e^{\frac{N}{2}\sum_{a=1}^{n}\hat{m}_a m_a + \frac{N}{2}\sum_{a<b}\hat{Q}_{ab}Q_{ab} + N\frac{\beta}{2}\sum_{a=1}^{n} m_a^2 - \frac{P-1}{2}log|I-\beta Q|}\cdot \tag{16}$$
$$\cdot e^{N\log\left(\mathbb{E}_x \sum_{\sigma\in\{-1,1\}^{N\times n}} e^{-\frac{1}{2}\sum_{a=1}^{n}\hat{m}_a x\sigma^a - \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}\sigma^a\sigma^b}\right)}.$$

Now, 16 is just a fixed-dimension integral of an energy depending on N. We can now apply the saddle point approximation and simply look for the maximum of the exponent.

Lemma 4.1 In the limit $N \to \infty$, $f^{(n)} \to extr_{m,\hat{m},Q,\hat{Q}} \Phi^{(n)}(m,\hat{m},Q,\hat{Q})$, where

$$\Phi^{(n)}(m,\hat{m},Q,\hat{Q}) = \frac{1}{2}\sum_{a,b} \hat{m}_a m_a + \frac{1}{2}\sum_{a<b} \hat{Q}_{ab} Q_{ab} + \frac{\beta}{2}\sum_a m_a^2 - \frac{\alpha}{2}\log|I-\beta Q| + \log\left(\mathbb{E}_x \sum_{\sigma\in\{-1,1\}^n} e^{-\frac{1}{2}\sum_{a=1}^n \hat{m}_a \sigma^a x - \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}\sigma^a\sigma^b}\right). \tag{17}$$

Proof. From our equation 16, we define:

$$g(x) = \sum_{i=1}^N e^{-\frac{1}{2}\sum_a \hat{m}_a x_i^{(1)}\cdot\sigma_i^a - \frac{1}{2}\sum_a \hat{Q}_{ab}\sigma_i^a\cdot\sigma_i^b}.$$

As x_i are i.i.d, we can drop the index and consider only x ($x \in \{-1,1\}$ is a single entry of $x^{(1)}$).

For the same reason:

$$\mathbb{E}_{x^{(1)}} \prod_{i=1}^N g(x_i^{(1)}) = (\mathbb{E}_x g(x))^N = e^{N\log\mathbb{E}_x g(x)}.$$

On the other hand, as $\alpha = \frac{P}{N}$,

$$\frac{P-1}{2}\log|I-\beta Q| = N\frac{\alpha}{2}\log|I-\beta Q| + O(\frac{1}{N}).$$

We can forget about $O(\frac{1}{N})$, since we will apply the limit $N \to \infty$. Then, in our expression (11):

$$\frac{P-1}{2}\log|I-\beta Q| = N\frac{\alpha}{2}\log|I-\beta Q|.$$

Note that we find that all the terms in the exponential expression of (16) are multiplied by N, so we are led to define

$$\Phi^{(n)}(m,\hat{m},Q,\hat{Q}) = \frac{1}{2}\sum_{a,b}\hat{m}_a m_a + \frac{1}{2}\sum_{a<b}\hat{Q}_{ab}Q_{ab} + \frac{\beta}{2}\sum_a m_a^2 - \frac{\alpha}{2}\log|I-\beta Q| + \log(\mathbb{E}_x g(x)).$$

Up to this point we have

$$f^{(n)} = \frac{1}{N} \int \left[\prod_{a=1}^{N} \frac{dm_a d\hat{m}_a}{2\pi/N} \right] \left[\prod_{1 \le a < b \le n} \frac{dQ_{ab} d\hat{Q}_{ab}}{2\pi/N} \right] e^{N\Phi^{(n)}(m,\hat{m},Q,\hat{Q})}.$$

Now we define:

$$d\mu = \left[\prod_{a=1}^{N} \frac{dm_a d\hat{m}_a}{2\pi/N} \right] \left[\prod_{1 \le a < b \le n} \frac{dQ_{ab} d\hat{Q}_{ab}}{2\pi/N} \right].$$

That is,

$$f^{(n)} = \frac{1}{N} \log \int d\mu e^{N\Phi^{(n)}(m,\hat{m},Q,\hat{Q})}.$$

Applying the limit $N \to \infty$ and following Laplace's method,

$$\int d\mu e^{N\Phi} \approx e^{N\Phi(\star)},$$

where $\Phi(\star)$ is the value of Φ at the stationary point $\star$. Therefore,

$$\lim_{N \to \infty} f^{(n)} = extr_{m,\hat{m},Q,\hat{Q}} \Phi^{(n)}(m, \hat{m}, Q, \hat{Q}).$$

Where *extr* means maximizing or minimizing (taking the derivative and setting to 0 specifically), which compacts the notation used with $\star$.

□

Note: We seek a maximum because as $N \to \infty$, terms with larger exponent overwhelm terms with smaller exponent, so the sum effectively collapses onto the terms with the largest exponent. Therefore, we will need the maximum of Φ.

We don't know the optimum of 17 and have little hope of obtaining an expression for general n that would allow us to take the $n \to 0$ limit required by the replica trick. Thus, we insert a very simple ansatz for the optimum, known as the **replica symmetric ansatz**.

This consists in assuming that all replica parameters are symmetric with respect to interchange of the replicas $a = 1, \ldots, n$. That is

$$m_a = m, \quad \hat{m}_a = \hat{m}, \quad Q_{ab} = q, \quad \hat{Q}_{ab} = \hat{q},$$

for all $1 \le a < b \le n$, where $m, \hat{m}, q, \hat{q}$ are real constants.
Then, we can define the replica symmetric potential

$$\begin{aligned} \Phi_{\mathrm{RS}}^{(n)}(m, \hat{m}, q, \hat{q}) = \frac{n}{2} \hat{m}\, m + \frac{n^2 - n}{4} \hat{q}\, q + \frac{\beta}{2} n\, m^2 - \frac{\alpha}{2} \log \left| I - \beta Q_{\mathrm{RS}} \right| + \\ + \log \left(\mathbb{E}_x \sum_{\sigma \in \{-1,1\}^n} \exp \left(-\tfrac{1}{2} \hat{m} \sum_{a=1}^{n} \sigma^a \cdot x - \tfrac{1}{2} \hat{q} \sum_{1 \le a < b \le n} \sigma^a \cdot \sigma^b \right) \right), \end{aligned} \tag{18}$$

where Q_{RS} has diagonal 1 and all other entries are equal to q.

This last expression is obtained by rewriting some terms of 17,

$$\frac{1}{2}\sum_{a=1}^{n}\hat{m}_a m_a = \frac{1}{2}\sum_{a=1}^{n}\hat{m}m = \frac{n}{2}\,\hat{m}\,m,$$

$$\frac{1}{2}\sum_{a<b}\hat{Q}_{ab}Q_{ab} = \frac{1}{2}\cdot\frac{n(n-1)}{2}\,\hat{q}\,q = \frac{n^2-n}{4}\,\hat{q}\,q,$$

$$\frac{\beta}{2}\sum_{a=1}^{n}m_a^2 = \frac{\beta}{2}\,n\,m^2,$$

and substituting them to obtain 18.

By calculating the determinant $|I-\beta Q_{RS}|$ and using the identity

$$\mathbb{E}_{z\sim\mathcal{N}(0,1)}[e^{zc}] = e^{\frac{1}{2}c^2},$$

we can simplify the last two terms of 18 leading to

$$\begin{aligned}\Phi_{RS}^{(n)}(m,\hat{m},q,\hat{q}) := \frac{n}{2}\hat{m}m + \frac{1}{2}\frac{n^2-n}{2}\hat{q}q + \frac{\beta}{2}nm^2 \\ - n\frac{\alpha}{2}\log\left(1-\beta\left(1-q\right)\right) - \frac{\alpha}{2}\log\left(1-n\frac{\beta q}{\left(1-\beta\left(1-q\right)\right)}\right) \\ + \log\left(\mathbb{E}_{z,x}\left[\left(\sum_{\sigma\in\{-1,1\}} e^{-\frac{1}{2}\hat{m}\sigma x+\frac{1}{4}\hat{q}+z\sqrt{-\frac{1}{2}\hat{q}}\sigma}\right)^n\right]\right).\end{aligned} \tag{19}$$

(see details in 4.1.6.)

If we combine 19 with our expression for $f^{(n)}$, we get

$$f_{RS} = \lim_{n\to 0} extr_{m,\hat{m},Q,\hat{Q}}\frac{1}{n}\Phi_{RS}^{(n)}(m,\hat{m},q,\hat{q}).$$

Remember that the replica trick requires us to take the number of replicas to 0, so we will need to calculate the limit when $n\to 0$.

Lemma 4.2. Assuming we can bring the $n\to 0$ limit into the extremization,

$$\lim_{n\to 0} extr_{m,\hat{m},q,\hat{q}}\frac{1}{n}\Phi^{(n)}(m,\hat{m},q,\hat{q}) = extr_{m,\hat{m},q,\hat{q}}\Phi^{(n)}(m,\hat{m},q,\hat{q}),$$

where,

$$\Phi^{(n)}(m,\hat{m},q,\hat{q}) = \frac{1}{2}\hat{m}m - \frac{1}{4}\hat{q}q + \frac{\beta}{2}m^2 - \frac{\alpha}{2}\log(1-\beta(1-q)) + \frac{\alpha}{2}\frac{\beta q}{1-\beta(1-q)}$$

$$+\frac{\hat{q}}{4} + \mathbb{E}_{x,z}\left[\log\left(\cosh\left(-\frac{1}{2}\hat{m}x + z\sqrt{-\frac{\hat{q}}{2}}\right)\right)\right] + \log 2. \tag{20}$$

Proof. We have

$$\frac{1}{n}\Phi^{(n)}_{RS} = \frac{1}{2}\hat{m}m + \left(\frac{n}{4}-\frac{1}{4}\right)\hat{q}q + \frac{\beta}{2}m^2 - \frac{\alpha}{2}\log(1-\beta(1-q)) - \frac{\alpha}{2n}\log\left(1-\frac{n\beta q}{(1-\beta(1-q))}\right) +$$

$$\frac{1}{n}\log\left(\mathbb{E}_{z,x}\left[\left(\sum_{\sigma\in\{-1,1\}} e^{-\frac{1}{2}\hat{m}\sigma x + \frac{1}{4}\hat{q} + z\sqrt{-\frac{\hat{q}}{2}}\sigma}\right)^n\right]\right).$$

Using the fact that $\log(1+nx) = nx + O(n^2)$, the second logarithmic term can be written

$$-\frac{\alpha}{2n}\log\left(1-\frac{n\beta q}{(1-\beta(1-q))}\right) = \frac{\alpha}{2}\frac{\beta q}{1-\beta(1-q)} + O(n).$$

Next, writing H^n for the quantity in the expectation of (20), we can apply the replica trick in reverse, ie.

$$\frac{1}{n}\log\left(\mathbb{E}_{z,x}\left[H^n\right]\right) \to \mathbb{E}_{z,x}\log H.$$

Finally, note that H can be written

$$H(x,z) = \sum_{\sigma\in\{-1,1\}} e^{-\frac{1}{2}\hat{m}\sigma x + \frac{1}{4}\hat{q} + z\sqrt{-\frac{\hat{q}}{2}}\sigma} = 2e^{\frac{\hat{q}}{4}}\cosh\left(-\frac{1}{2}\hat{m}x + z\sqrt{-\frac{\hat{q}}{2}}\right)$$

Substituting back into (20) and computing limits of the other terms, we find

$$\lim_{n\to 0}\frac{1}{n}\Phi^{(n)}(m,\hat{m},q,\hat{q}) = \frac{1}{2}\hat{m}m - \frac{1}{4}\hat{q}q + \frac{\beta}{2}m^2 - \frac{\alpha}{2}\log(1-\beta(1-q)) + \frac{\alpha}{2}\frac{\beta q}{1-\beta(1-q)}$$

$$+\frac{\hat{q}}{4} + \mathbb{E}_{x,z}\left[\log\left(\cosh\left(-\frac{1}{2}\hat{m}x + z\sqrt{-\frac{\hat{q}}{2}}\right)\right)\right] + \log 2.$$

□

Note that we have finally arrived at the potential 6, whose critical points describe the network's behavior. What remains is finding the saddle-point equations.

To do so, we differentiate 20 with respect to $m, q, \hat{m}$ and $\hat{q}$ and equate to 0, leading to:

$$\begin{cases} \hat{m} = -2\beta m, \\ \hat{q} = -\frac{2\alpha\beta^2 q}{(1-\beta(1-q))^2}, \\ m = \mathbb{E}_{z,x}\left[x \tanh\left(-\frac{1}{2}\hat{m}x + z\sqrt{-\frac{1}{2}\hat{q}}\right)\right], \\ q = 1 - \frac{1}{\sqrt{-\frac{1}{2}\hat{q}}}\mathbb{E}_{z,x}\left[\tanh\left(-\frac{1}{2}\hat{m}x + z\sqrt{-\frac{1}{2}\hat{q}}\right)z\right]. \end{cases}$$

Substituting the expressions for $\hat{m}$ and $\hat{q}$ into the m and q equations and then using **Stein's Lemma**,

$$\mathbb{E}_{z\sim\mathcal{N}(0,1)}[zf(z)] = \mathbb{E}_{z\sim\mathcal{N}(0,1)}\left[\frac{df}{dz}(z)\right],$$

we obtain an expression for m and q that are expectations that depend on variables x and z. Using that the variable x in the expectation takes values in -1, 1 with probability $\frac{1}{2}$, we can finally simplify the saddle point equations to

$$m = \mathbb{E}_z\left[\tanh\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]$$
$$q = \mathbb{E}_z\left[\tanh^2\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right],$$

which is exactly what we were looking for.
(See details in 4.1.7)

□

4.1.4 Calculus of the integral

We want to calculate

$$\int e^{-\frac{1}{2}u^\top Q^{-1}u}e^{\beta\frac{1}{2}u^\top u}du.$$

Note that can re-represent what is inside of the integral as a single exponential with exponent $-\frac{1}{2}u^T(Q^{-1} - \beta I)u$. For the Gaussian integral to converge, the matrix $(Q^{-1} - \beta I)^{-1}$ must be symmetric definite positive. We will assume it to proceed with the calculation. Using that

$$\sqrt{|2\pi\Sigma|} = \int e^{-\frac{1}{2}x^\top\Sigma^{-1}x}dx$$

for any symmetric positive definite matrix Σ, we have that the integral equals $\sqrt{\frac{|2\pi I|}{|Q^{-1}-\beta I|}}$.

Therefore,

$$\mathbb{E}_{u\sim N(0,Q)} e^{\beta \frac{1}{2} u^T u} = \frac{1}{\sqrt{|2\pi Q|}} \sqrt{\frac{|2\pi I|}{|Q^{-1} - \beta I|}} = \frac{1}{\sqrt{|Q||Q^{-1} - \beta I|}} = |I - \beta Q|^{-\frac{1}{2}}.$$

Plugging it into 13 we obtain

$$f^{(n)} = \frac{1}{N} \log \mathbb{E}_{x^{(1)}} \sum_{\sigma \in \{-1,1\}^{N\times n}} e^{\beta \frac{N}{2} \sum_{a=1}^{N} m_a^2} (|I - \beta Q|^{-\frac{1}{2}})^{P-1}.$$

We can write $|I - \beta Q|^{-\frac{1}{2}}$ as $e^{\log\left(|I-\beta Q|^{-\frac{1}{2}}\right)} = e^{-\frac{1}{2}\log(|I-\beta Q|)}$ and easily get 14.

4.1.5 Restrictions among a, b in overlap variables

On one hand, restrictions at the extremes of the inequality are trivial, as $\sigma' s$ index goes from 1 to n.

On the other hand, we note that the matrix Q is symmetric:

$$Q_{ab} = \frac{1}{N}\sigma_a\sigma_b = \frac{1}{N}\sigma_b\sigma_a = Q_{ba}$$

For this reason we do not need to include both inequalities $a < b$ and $b < a$, as it would count the same value twice.

Finally, if we consider the case $a = b$, we can see that:

$$Q_{aa} = \frac{1}{N}\sigma_a\sigma_a = 1$$

Therefore, we do not need to impose $a = b$, just $a < b$.

4.1.6 Symmetry ansatz

First we calculate $|I - \beta Q_{RS}|$.

We use that

$$Q_{RS} = (1-q)I + q\,\mathbf{1}\mathbf{1}^\top \;\Rightarrow\; I - \beta Q_{RS} = (1 - \beta(1-q))I - \beta q\,\mathbf{1}\mathbf{1}^\top.$$

Thus, we have a matrix of the form $aI + b\,\mathbf{1}\mathbf{1}^\top$.

The matrix $\mathbf{1}\mathbf{1}^\top$ has eigenvalues n (multiplicity 1) and 0 (multiplicity $n-1$), so $aI + b\,\mathbf{1}\mathbf{1}^\top$ has eigenvalues $\lambda_1 = a + bn$ (one) and $\lambda_{2\ldots n} = a$ (n-1). Therefore, it's determinant is $|I - \beta Q_{RS}| = a^{n-1}(a + bn)$.

Hence,

$$|I - \beta Q_{RS}| = (1 - \beta(1-q))^{n-1}\big(1 - \beta(1-q) - n\beta q\big) = (1-\beta(1-q))^n \left(1 - \frac{n\beta q}{1 - \beta(1-q)}\right)$$

Now we focus on simplifying the last term of 18.

$$\mathcal{L} := \log\left(\mathbb{E}_x \sum_{\boldsymbol{\sigma}\in\{\pm1\}^n} \exp\Big[-\tfrac{1}{2}\hat{m}\sum_{a=1}^{n}\sigma_a\cdot x - \tfrac{1}{2}\hat{q}\sum_{1\le a<b\le n}\sigma_a\cdot\sigma_b\Big]\right).$$

First, we use the algebraic identity

$$\sum_{1\le a<b\le n}\sigma_a\cdot\sigma_b = \tfrac{1}{2}\Big(\Big(\sum_{a=1}^{n}\sigma_a\Big)^2 - \sum_{a=1}^{n}\sigma_a^2\Big) = \tfrac{1}{2}\Big(\Big(\sum_a\sigma_a\Big)^2 - n\Big),$$

since $\sigma_a^2 = 1$.

Substituting this into the exponent gives:

$$-\tfrac{1}{2}\hat{q}\sum_{a<b}\sigma_a\cdot\sigma_b = -\tfrac{1}{4}\hat{q}\Big(\sum_a\sigma_a\Big)^2 + \tfrac{1}{4}\hat{q}\,n.$$

Therefore, the term inside the logarithm can be written as:

$$\mathcal{L} = \log\left(\mathbb{E}_x\sum_{\boldsymbol{\sigma}}\exp\Big[-\tfrac{1}{2}\hat{m}\sum_a\sigma_a\cdot x - \tfrac{1}{4}\hat{q}\big(\sum_a\sigma_a\big)^2 + \tfrac{1}{4}\hat{q}\,n\Big]\right)$$

Now we apply our identity! For any number S and constant A, we have:

$$e^{\frac{1}{2}AS^2} = \mathbb{E}_{z\sim\mathcal{N}(0,1)}[\,e^{z\sqrt{A}\,S}\,],$$

because $\mathbb{E}_z[e^{zc}] = e^{c^2/2}$.

We set $S = \sum_a\sigma_a$ and note that

$$-\tfrac{1}{4}\hat{q}(\sum_a\sigma_a)^2 = \tfrac{1}{2}\Big(-\tfrac{\hat{q}}{2}\Big)(\sum_a\sigma_a)^2,$$

so that $A = -\frac{\hat{q}}{2}$.

Then,

$$\exp\Big(-\tfrac{1}{4}\hat{q}(\sum_a\sigma_a)^2\Big) = \mathbb{E}_z\Big[\exp\Big(z\sqrt{-\tfrac{\hat{q}}{2}}\sum_a\sigma_a\Big)\Big].$$

Inserting this representation back into the sum over σ, we obtain:

$$\sum_{\boldsymbol{\sigma}\in\{\pm1\}^n}\exp\Big[-\tfrac{1}{2}\hat{m}\sum_a\sigma_a\cdot x - \tfrac{1}{4}\hat{q}(\sum_a\sigma_a)^2\Big] = \mathbb{E}_z\left[\Big(\sum_{\sigma\in\{\pm1\}} e^{-\frac{1}{2}\hat{m}\sigma x + \frac{1}{4}\hat{q} + z\sqrt{-\frac{\hat{q}}{2}}\sigma}\Big)^n\right].$$

Finally, we can write:

$$\mathcal{L} = \log\left(\mathbb{E}_{z,x}\Big[\Big(\sum_{\sigma\in\{\pm1\}} e^{-\frac{1}{2}\hat{m}\sigma x + \frac{1}{4}\hat{q} + z\sqrt{-\frac{\hat{q}}{2}}\sigma}\Big)^n\Big]\right)$$

Now we put everything together and we can finally write 19.

4.1.7 Differentiation

If we differentiate 20 with respect to m and q, we obtain the following:

$$\frac{\partial \Phi_{RS}(m,\hat{m},q,\hat{q})}{\partial m} = \frac{1}{2}\hat{m} + \beta m$$
$$\frac{\partial \Phi_{RS}(m,\hat{m},q,\hat{q})}{\partial q} = -\frac{1}{4}\hat{q} + \frac{\alpha\beta}{2}\frac{-\beta q}{(1-\beta(1-q))^2}$$

By equating these expressions to 0 and isolating, we get:

$$\frac{1}{2}\hat{m} + \beta m = 0 \iff \hat{m} = -2\beta m$$

$$-\frac{1}{4}\hat{q} + \frac{\alpha\beta}{2}\frac{-\beta q}{(1-\beta(1-q))^2} = 0 \iff \hat{q} = -\frac{2\alpha\beta^2 q}{(1-\beta(1-q))^2}$$

To differentiate with respect to $\hat{m}$ and $\hat{q}$, we define $\square = -\frac{1}{2}\hat{m}x + z\sqrt{-\frac{1}{2}\hat{q}}$. Then

$$\frac{\partial \square}{\partial \hat{m}} = -\frac{1}{2}x,$$
$$\frac{\partial \square}{\partial \hat{q}} = \frac{z}{2\sqrt{-\frac{1}{2}\hat{q}}}\Big(-\frac{1}{2}\Big).$$

Now,

$$\frac{\partial \Phi_{RS}(m,\hat{m},q,\hat{q})}{\partial \hat{m}} = \frac{1}{2}m + \mathbb{E}_{z,x}\Bigg[\frac{\partial}{\partial \hat{m}}\log\cosh(\square)\Bigg],$$

$$\frac{\partial \Phi_{RS}(m,\hat{m},q,\hat{q})}{\partial \hat{q}} = -\frac{1}{4}q + \frac{1}{4} + \mathbb{E}_{z,x}\Bigg[\frac{\partial}{\partial \hat{q}}\log\cosh(\square)\Bigg].$$

As

$$\frac{\partial}{\partial \hat{m}}\log\cosh(\square) = \frac{1}{\cosh(\square)}\sinh(\square)\Big(-\frac{1}{2}x\Big) = -\frac{1}{2}x\tanh(\square),$$

$$\frac{\partial}{\partial \hat{q}}\log\cosh(\square) = \tanh(\square)\frac{z}{2\sqrt{-\frac{1}{2}\hat{q}}}\Big(-\frac{1}{2}\Big),$$

we finally have that

$$\frac{\partial \Phi_{RS}(m,\hat{m},q,\hat{q})}{\partial \hat{m}} = \frac{1}{2}m + -\frac{1}{2}\mathbb{E}_{z,x}\Big[x\tanh(\square)\Big],$$

$$\frac{\partial \Phi_{RS}(m,\hat{m},q,\hat{q})}{\partial \hat{q}} = -\frac{1}{4}q + \frac{1}{4} - \frac{1}{4\sqrt{-\frac{1}{2}\hat{q}}}\mathbb{E}_{z,x}\left[\tanh(\square)z\right].$$

As we have done before, by equating these expressions to 0 and isolating, we obtain the following:

$$\frac{1}{2}m + -\frac{1}{2}\mathbb{E}_{z,x}\left[x\tanh(\square)\right] = 0 \iff m = \mathbb{E}_{z,x}\left[x\tanh(\square)\right],$$

$$-\frac{1}{4}q + \frac{1}{4} - \frac{1}{4\sqrt{-\frac{1}{2}\hat{q}}}\mathbb{E}_{z,x}\left[\tanh(\square)z\right] = 0 \iff$$
$$q = 1 - \frac{1}{\sqrt{-\frac{1}{2}\hat{q}}}\mathbb{E}_{z,x}\left[\tanh(\square)z\right].$$

Putting it all together, we obtain a system of 4 equations that completely characterizes the overlap variables:

$$\begin{cases} \hat{m} = -2\beta m \\ \hat{q} = -\frac{2\alpha\beta^2 q}{(1-\beta(1-q))^2} \\ m = \mathbb{E}_{z,x}\left[x\tanh(\square)\right] \\ q = 1 - \frac{1}{\sqrt{-\frac{1}{2}\hat{q}}}\mathbb{E}_{z,x}\left[\tanh(\square)z\right] \end{cases}$$

Now we will substitute $\hat{m}$ and $\hat{q}$ in the m and q equations. First, we have $-\frac{1}{2}\hat{m}x = \beta m x$ and $\sqrt{-\frac{1}{2}\hat{q}} = \sqrt{\frac{\alpha\beta^2 q}{(1-\beta(1-q))^2}} = \beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}$. Considering these, we have

$$m = \mathbb{E}_{z,x}\left[x\tanh\left(\beta m x + z\beta\frac{\sqrt{\alpha q}}{(1-\beta(1-q))}\right)\right] = \mathbb{E}_{z,x}\left[x\tanh\left(\beta(mx + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right],$$

$$q = 1 - \frac{1}{\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}}\mathbb{E}_{z,x}\left[z\tanh\left(\beta(mx + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right].$$

As $z \sim \mathcal{N}(0,1)$, we can use Stein's lemma:

$$\mathbb{E}_{z,x}\left[z\tanh\left(\beta(mx + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right] = \mathbb{E}_{z,x}\left[\frac{\partial}{\partial z}\tanh\left(\beta(mx + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right] =$$

$$\mathbb{E}_{z,x}\left[\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}\left(1-\tanh^2\left(\beta(mx+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right)\right].$$

Therefore, we have

$$q=1-\frac{1}{\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}}\mathbb{E}_{z,x}\left[\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}\left(1-\tanh^2\left(\beta(mx+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right)\right]=$$

$$1-\frac{1}{\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}}\mathbb{E}_{z,x}\left[\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}\right]+\frac{1}{\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}}\mathbb{E}_{z,x}\left[\beta\frac{\sqrt{\alpha q}}{1-\beta(1-q)}\tanh^2\left(\beta(mx+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]=$$

$$\mathbb{E}_{z,x}\left[\tanh^2\left(\beta(mx+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right].$$

With all this, we obtain

$$m=\mathbb{E}_{z,x}\left[x\tanh\left(\beta(mx+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right],$$

$$q=\mathbb{E}_{z,x}\left[\tanh^2\left(\beta(mx+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right].$$

Now, we average over the variable x that can take values +1 and -1 with equal probability:

$$q=\frac{1}{2}\left(\mathbb{E}_z\left[\tanh^2\left(\beta(1m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]+\mathbb{E}_z\left[\tanh^2\left(\beta(m(-1)+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]\right)=$$

$$\mathbb{E}_z\left[\tanh^2\left(\beta(m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right].$$

In the last step, we have used that $\tanh^2 x$ is an even function, so the averages for x = 1 and x = -1 are the same.

$$m=\frac{1}{2}\left(\mathbb{E}_z\left[\tanh\left(\beta(m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]+\mathbb{E}_z\left[(-1)\tanh\left(\beta(m(-1)+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]\right)=$$

$$\frac{1}{2}\left(\mathbb{E}_z\left[\tanh\left(\beta(m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]-\mathbb{E}_z\left[\tanh\left(\beta(-m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]\right)=$$

$$\frac{1}{2}\left(\mathbb{E}_z\left[\tanh\left(\beta(m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]+\mathbb{E}_z\left[\tanh\left(\beta(m-z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]\right)$$

In the last step, we have used that $\tanh x$ is an odd function, so $\mathbb{E}_z\left[\tanh\left(\beta(-m+z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]=-\mathbb{E}_z\left[\tanh\left(\beta(m-z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]$.

Due to the symmetry of the distribution $\mathcal{N}(0,1)$, we can say that $\mathbb{E}_z\left[\tanh\left(\beta(m - z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right] = \mathbb{E}_z\left[\tanh\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]$. Therefore,

$$m = \mathbb{E}_z\left[\tanh\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right].$$

We have simplified the saddle point equations to

$$m = \mathbb{E}_z\left[\tanh\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]$$
$$q = \mathbb{E}_z\left[\tanh^2\left(\beta(m + z\frac{\sqrt{\alpha q}}{(1-\beta(1-q))})\right)\right]$$

,
which is exactly what we wanted to see.

4.1.8 Delta function limit

Using *Desmos*, we can infer that the limit of the expression

$$\beta \operatorname{sech}^2(\beta x) \qquad \beta \to \infty,$$

is of the form $a\,\delta(x)$. To determine the exact value of the prefactor a, we apply the definition of the Dirac delta function.

$$\boxed{\int_{-\infty}^{\infty} f(x)\,\delta(x-a)\,dx = f(a)}.$$

$$\int_{-\infty}^{\infty} \beta \operatorname{sech}^2(\beta x)\,dx = \int_{-\infty}^{\infty} \operatorname{sech}^2(y)\ dy = \tanh(y)\Big|_{-\infty}^{\infty} = 2 \Rightarrow \boxed{a=2}.$$

We can then conclude that $\beta \operatorname{sech}^2(\beta x) \to 2\delta(x)$.

4.2 Perceptron

4.2.1 Justification of "flip" trick

In the first place, we will define an "auxiliary" pattern $\tilde{x}^{(\mu)} = y^{(\mu)}x^{(\mu)}$. We want to show that the original condition $y^{(\mu)}\frac{w\cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$ can be transformed to $\frac{w\cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$. We need to check the definition for the cases $y^{(\mu)} \in \{-1, 1\}$.

- Case 1: $y^{(\mu)} = +1$
  - Original condition: $(+1)\frac{w\cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa \implies \frac{w\cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$.

  - New pattern: $\tilde{x}^{(\mu)} = (+1)x^{(\mu)} = x^{(\mu)}$.
  - New condition: $\frac{w \cdot \tilde{x}^{(\mu)}}{\sqrt{N}} \geq \kappa \implies \frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$.
  - The conditions are identical.

- Case 2: $y^{(\mu)} = -1$
  - Original condition: $(-1)\frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa \implies -\frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$.
  - New pattern: $\tilde{x}^{(\mu)} = (-1)x^{(\mu)} = -x^{(\mu)}$.
  - New condition: $\frac{w \cdot \tilde{x}^{(\mu)}}{\sqrt{N}} \geq \kappa \implies \frac{w \cdot (-x^{(\mu)})}{\sqrt{N}} \geq \kappa \iff -\frac{w \cdot x^{(\mu)}}{\sqrt{N}} \geq \kappa$.
  - The conditions are again identical.

Consequently, we can see that the volume defined by the original condition is the same as the one that uses the pattern $\tilde{x}^{(\mu)} = y^{(\mu)}x^{(\mu)}$. However, this "flipping trick" only works if the new patterns $\{\tilde{x}^{(\mu)}\}$ have the same statistical properties as the original ones $\{x^{(\mu)}\}$.

We are mainly interested in the normalized logarithm

$$\mathcal{V} = \mathbb{E}_x \frac{1}{N} log\ V$$

The $\mathbb{E}_x$ denotes an average over the "disorder", which is the set of all patterns $\{-1, 1\}$.
Taking into account that the original patterns are independently sampled from a standard normal distribution (i.e. $x^{(\mu)} \sim \mathcal{N}(0, I_N)$), we observe that the symmetry property will be worthy of consideration. First, its probability distribution f is centered at 0 and satisfies $f(-x) = f(x)$. In addition, we note that a random variable x selected from $\mathcal{N}(0, I_N)$ has the same probability distribution as the variable $-x$.

Now, remember that the new patterns $\{\tilde{x}^{(\mu)}\}$ are defined by taking the original patterns $\{x^{(\mu)}\}$ and, for each μ where $y^{(\mu)} = -1$, we "flip" the pattern from $x^{(\mu)}$ to $-x^{(\mu)}$. Using the properties mentioned above, we can see that $x^{(\mu)}$ and $-x^{(\mu)}$ are statistically identical.

Therefore, when computing the average over all possible random patterns, the two averages are identical:

$$\mathbb{E}_{\{x^{(\mu)}\}}\left[\log V\left(\{x^{(\mu)}\}, \{y^{(\mu)}\}\right)\right] = \mathbb{E}_{\{\tilde{x}^{(\mu)}\}}\left[\log V\left(\{\tilde{x}^{(\mu)}\}, \{+1\}\right)\right]$$

So, since the distributions are the same, we can compute the average using the simplified form of V. After renaming $\tilde{x}$ back to x, we get the desired expression.

4.2.2 Theorem 2 Proof

We will start from the expression obtained from the replica trick, that involves n replicated systems. We will now simplify it by averaging over the disorder (the random patterns).

Assume $x^{(\mu)}$ are i.i.d. patterns, then the expectation factorizes as:

$$\mathbb{E}_{\mathbf{x}}\left[\prod_{\mu=1}^{P}\prod_{a=1}^{n}\Theta\left(\frac{\mathbf{w}^a\cdot\mathbf{x}^{(\mu)}}{\sqrt{N}}-\kappa\right)\right]=\left[\Psi(\mathbf{w}^1,\ldots,\mathbf{w}^n)\right]^P,$$

where

$$\Psi(\mathbf{w}^1,\ldots,\mathbf{w}^n)=\mathbb{E}_{\mathbf{x}}\prod_{a=1}^{n}\Theta\left(\frac{\mathbf{w}^a\cdot\mathbf{x}}{\sqrt{N}}-\kappa\right).$$

Substituting this into our expression 10, we obtain:

$$\mathcal{V}=\lim_{n\to 0}\frac{1}{nN}\log\int\left[\prod_{a=1}^{n}d\mathbf{w}^a\right]\left[\Psi(\mathbf{w}^1,\ldots,\mathbf{w}^n)\right]^P.$$

Finally, since $P=\alpha N$, we get

$$\mathcal{V}=\lim_{n\to 0}\frac{1}{nN}\log\int\left[\prod_{a=1}^{n}dw^a\right]e^{N\alpha\log\Psi(w^1,\ldots,w^n)}. \tag{21}$$

We now have an expression that depends on α and contains an exponential term, which will be much more convenient for subsequent analytical derivations. As we did in theorem 1 proof, we will introduce some order parameters.
The volume $\mathcal{V}$ now depends on the weight vectors through the function Ψ, which is

$$\Psi(w^1,\ldots,w^n)=\mathbb{E}_x\prod_{a=1}^{n}\Theta\left(\frac{w^a\cdot x}{\sqrt{N}}-\kappa\right).$$

We observe that this function depends on the vectors w^a only through their products with the random patterns,

$$\lambda^a=\frac{w^a\cdot x}{\sqrt{N}}.$$

Their joint distribution is characterized by their covariances as it is a Gaussian random variable, and we can do as follows:

$$\mathbb{E}\left[\lambda^a\lambda^b\right]=\mathbb{E}\left[\frac{w^a\cdot x}{\sqrt{N}}\frac{w^b\cdot x}{\sqrt{N}}\right]=\frac{1}{N}\sum_{i,j}w_i^a w_j^b\mathbb{E}[x_ix_j]=\frac{1}{N}\sum_i w_i^a w_i^b,$$

as $\mathbb{E}[x_ix_j]=\delta_{i,j}$. This leads us to define the order parameter

$$Q_{ab} = \frac{1}{N} w^a w^b,$$

for $a, b = 1, ..., n$.

In fact, this is the overlap between the different replicated weight vectors. Note that the weights are constrained to the sphere $||w||^2 = N$. So, for all $a = 1, ..., n$:

$$Q_{aa} = \frac{1}{N} w^a w^a = 1.$$

Note also that the mean of λ^a is

$$\mathbb{E}[\lambda^a] = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} w_i^a \mathbb{E}[x_i] = 0.$$

Therefore, the expectation over x can be performed knowing only the matrix Q_{ab} resulting in

$$\Psi(w^1, ..., w^n) = \Psi(Q),$$

where $\Psi(Q)$ is defined as the Gaussian integral

$$\Psi(Q) = \int d^n \lambda p(\lambda) \prod_{a=1}^{n} \Theta(\lambda^a - \kappa),$$

and $p(\lambda)$ is the multivariate Gaussian density with 0 mean and covariance matrix Q. This probability density is:

$$p(\lambda) = \frac{1}{(2\pi)^{n/2}\sqrt{det(Q)}} exp\left(-\frac{1}{2}\sum_{a,b} \lambda^a (Q^{-1})_{ab} \lambda^b\right).$$

Therefore, we obtain

$$\Psi(Q) = \int \frac{d^n \lambda}{(2\pi)^{n/2}\sqrt{det(Q)}} exp\left(-\frac{1}{2}\sum_{a,b} \lambda^a (Q^{-1})_{ab} \lambda^b\right) \prod_{a=1}^{n} \Theta(\lambda^a - \kappa).$$

Finally, if we substitute these values into our volume, we get:

$$\begin{aligned}
\mathcal{V} &= \lim_{n \to 0} \frac{1}{nN} \log \int \left[\prod_{a=1}^{n} dw^a\right] \left[\Psi(w^1, ..., w^n)\right]^{\alpha N} \\
&= \lim_{n \to 0} \frac{1}{nN} \log \int \left[\prod_{a<b} dQ_{ab}\right] \exp\{(N[\alpha \log \Psi(Q) + G(Q)])\},
\end{aligned}$$

where

$$\exp(NG(Q)) = \int \left[\prod_{a=1}^{n} dw^a\right] \prod_{a<b} \delta(NQ_{ab} - w^a \cdot w^b).$$

We have arrived at an expression where the volume $\mathcal{V}$ depends on the order parameters Q_{ab}. These Q_{ab} are defined in terms of the weight vectors. To decouple them, we introduce delta functions to enforce these definitions, in a similar way as we did in Theorem 1 proof. This will give us

$$\prod_{a<b} \delta(NQ_{ab} - w^a \cdot w^b) = \int \left[\prod_{a<b} \frac{N d\hat{Q}_{ab}}{2\pi}\right] \exp\left(\frac{N}{2}\sum_{a<b} \hat{Q}_{ab}(Q_{ab} - \frac{1}{N} w^a \cdot w^b)\right).$$

We can now substitute this into the expression for $\exp(NG(Q))$:

$$\exp(NG(Q)) = \int \left[\prod_{a<b} \frac{d\hat{Q}_{ab}}{\frac{2\pi}{N}}\right] \int \left[\prod_{a=1}^{n} dw^a\right] \exp\left(\frac{N}{2}\sum_{a<b} \hat{Q}_{ab}\left(Q_{ab} - \frac{1}{N} w^a \cdot w^b\right)\right).$$

Putting it all together, we obtain the following expression:

$$\begin{aligned}\frac{1}{N}\log \mathbb{E}[V^n] =& \frac{1}{N}\log \int_{(S^{N-1})^n} \left[\prod_{a=1}^{n} dw_a\right] \int \left[\prod_{a<b} \frac{dQ_{ab}\, d\hat{Q}_{ab}}{2\pi/N}\right] \\ & e^{\frac{N}{2}\sum_{a<b} \hat{Q}_{ab}\left(Q_{ab} - \frac{1}{N} w_a \cdot w_b\right)} e^{\alpha N \log\left(\mathbb{E}_{\mathbf{u}\sim\mathcal{N}(0,Q)}\left[\prod_{a=1}^{n} \Theta(u_a - \kappa)\right]\right)}.\end{aligned}$$

In a general calculation of this type, the state variables are the variables that are integrated/summed over in the original free energy expression - so in the case of the perceptron, the state variables are the components of the weight vector w. Our objective is to bring the integration over w (the state variables) inside. Observe that the term $\mathbb{E}_{u\sim\mathcal{N}(0,Q)}$ is the one that we defined before as $\Psi(Q)$.

Now, we will swap the order of integration in order to achieve our goal. Note that we are allowed to do it because the function that we are integrating is non-negative. After swapping the order, we obtain the following expression:

$$\begin{aligned}\frac{1}{N}\log \mathbb{E}[V^n] =& \frac{1}{N}\log \int \left[\prod_{a<b} \frac{dQ_{ab}\, d\hat{Q}_{ab}}{2\pi/N}\right] \int_{(S^{N-1})^n} \left[\prod_{a=1}^{n} dw_a\right] \\ & e^{\frac{N}{2}\sum_{a<b} \hat{Q}_{ab}\left(Q_{ab} - \frac{1}{N} w_a \cdot w_b\right)} e^{\alpha N \log\left(\mathbb{E}_{\mathbf{u}\sim\mathcal{N}(0,Q)}\left[\prod_{a=1}^{n} \Theta(u_a - \kappa)\right]\right)}.\end{aligned} \tag{22}$$

Notice that we can write the first exponential term as follows:

$$e^{\frac{N}{2}\sum_{a<b} \hat{Q}_{ab}(Q_{ab} - \frac{1}{N} w_a \cdot w_b)} = e^{\frac{N}{2}\sum_{a<b} \hat{Q}_{ab} Q_{ab}} e^{-\frac{N}{2}\sum_{a<b} \hat{Q}_{ab} \frac{1}{N} w_a \cdot w_b}.$$

Then, we can rewrite (22) factorizing the integral and grouping by w-independent and w-dependent terms:

$$\frac{1}{N}\log \mathbb{E}[V^n] = \frac{1}{N}\log \int \left[\prod_{a<b} \frac{dQ_{ab}\, d\hat{Q}_{ab}}{2\pi/N}\right] e^{\frac{N}{2}\sum_{a<b}\hat{Q}_{ab}Q_{ab}} e^{\alpha N \log\left(\mathbb{E}_{\mathbf{u}\sim\mathcal{N}(0,Q)}\left[\prod_{a=1}^{n}\Theta(u_a-\kappa)\right]\right)}$$
$$\int_{(S^{N-1})^n} \left[\prod_{a=1}^{n} dw_a\right] e^{-\frac{N}{2}\sum_{a<b}\hat{Q}_{ab}\frac{1}{N}w_a\cdot w_b}. \tag{23}$$

Observe that the second term, the one that depends on w, is similar to $\Xi(\hat{Q})$. Considering that the integration domain $(S^{N-1})^n$ can be understood as integrating in $\mathbb{R}^{nN}$ and enforcing n-spherical constraints $\|w_a\| = \sqrt{N}$, we rewrite it in the second way using δ-functions:

$$\int_{(S^{N-1})^n} \left[\prod_{a=1}^{n} dw_a\right] e^{-\frac{N}{2}\sum_{a<b}\hat{Q}_{ab}\frac{1}{N}w_a\cdot w_b} =$$
$$\int \left[\prod_{a=1}^{n} dw_a \delta(\|w_a\|^2 - N)\right] e^{-\frac{N}{2}\sum_{a<b}\hat{Q}_{ab}\frac{1}{N}w_a\cdot w_b} := I_w(\hat{Q}).$$

Notice that $\Xi(\hat{Q}) = \frac{1}{N}\log I_w(\hat{Q})$. From this we can deduce that $I_w(\hat{Q}) = e^{N\Xi(\hat{Q})}$.
Finally, replacing in (23), we end up with:

$$\frac{1}{N}\log \mathbb{E}[\mathcal{V}^n] = \frac{1}{N}\log \int \left[\prod_{a<b} \frac{dQ_{ab}d\hat{Q_{ab}}}{2\pi/N}\right] e^{N\left[\frac{1}{2}\sum_{a<b}Q_{ab}\hat{Q}_{ab} + \alpha\log\Phi(Q) + \Xi(\hat{Q})\right]} \tag{24}$$

where

$$\Phi(Q) = \mathbb{E}_{u\sim\mathcal{N}(0,Q)}\left[\prod_{a=1}^{n}\ \Theta(u_a-\kappa)\right]$$
$$\Xi(\hat{Q}) = \frac{1}{N}\log \int \left[\prod_{a=1}^{n} dw_a\delta(||w_a||^2 - N)\right] e^{-\frac{1}{2}\sum_{a,b}\hat{Q}_{ab}w_a\cdot w_b}$$

Up to this point, our equation is just a fixed-dimension integral of an energy depending on N. We can now apply the saddle point approximation and simply look for the maximum of the exponent.

We can prove that

$$\Xi(\hat{Q}) \to max_{\Lambda}\left[-tr[\Lambda] - \frac{1}{2}\log\left|\frac{1}{2}\hat{Q} - 2\Lambda\right| + \frac{n}{2}\log 2\pi\right]$$

where Λ is a diagonal $n \times n$ matrix.
We define

$$f(Q, \hat{Q}, \Lambda) = \frac{1}{2} \sum_{a<b} Q_{ab} \hat{Q}_{ab} + \alpha \log \Phi(Q) + \Xi(\hat{Q})$$

Therefore, our function can be rewritten as

$$\frac{1}{N} \log \mathbb{E}[\mathcal{V}^n] = \frac{1}{N} \log \int \left[\prod_{a<b} \frac{dQ_{ab} d\hat{Q_{ab}}}{2\pi/N} \right] e^{Nf(Q,\hat{Q},\Lambda)}$$

We vary f with respect to each set of variables. First, we consider the variation with respect to Q_{ab} for $a \neq b$:

$$\frac{\partial f}{\partial Q_{ab}} = \frac{1}{2} \hat{Q}_{ab} + \alpha \frac{1}{\Psi(Q)} \frac{\partial \Psi(Q)}{\partial Q_{ab}} = 0$$

And we get that

$$\hat{Q}_{ab} = -2\alpha \frac{1}{\Psi(Q)} \frac{\partial \Psi(Q)}{\partial Q_{ab}}$$

Now we vary with respect to $\hat{Q}_{ab}$ for $a \neq b$:

$$\frac{\partial f}{\partial \hat{Q}_{ab}} = \frac{1}{2} Q_{ab} + \frac{\partial \Xi(\hat{Q})}{\partial \hat{Q}_{ab}} = 0$$

So:

$$Q_{ab} = -2 \frac{\partial \Xi(\hat{Q})}{\partial \hat{Q}_{ab}}$$

Finally, we vary with respect to Λ_{aa} since Λ is diagonal. Let $M = \frac{1}{2}\hat{Q} - 2\Lambda$.

$$\frac{\partial \Xi(\hat{Q})}{\partial \Lambda_{aa}} = \frac{\partial}{\partial \Lambda_{aa}} \left[-tr[\Lambda] - \frac{1}{2} \log |M| \right] = -1 - \frac{1}{2} Tr \left[M^{-1} \frac{\partial M}{\partial \Lambda_{aa}} \right] = -1 + M_{aa}^{-1} = 0$$

because

$$Tr \left[M^{-1} \frac{\partial M}{\partial \Lambda_{aa}} \right] = \sum_{b,c} M_{bc}^{-1} (-2\delta_{ba}\delta_{ca}) = -2M_{aa}^{-1}$$

Therefore, substituting in M we obtain the following.

$$\left[\left(\frac{1}{2}\hat{Q} - 2\Lambda \right)^{-1} \right]_{aa} = 1$$

Note here that for large N, the integral is dominated by the saddle point where f is stationary with respect to Q_{ab}, $\hat{Q}_{ab}$ and Λ. Thus,

$$\frac{1}{N}\log \mathbb{E}[\mathcal{V}^n] \to_{N\to\infty} extr_{Q,\hat{Q},\Lambda} f(Q,\hat{Q},\Lambda) \tag{25}$$

where

$$f(Q,\hat{Q},\Lambda) = \frac{1}{2}\sum_{a<b} Q_{ab}\hat{Q}_{ab} + \alpha \log \Phi(Q) - \mathrm{tr}[\Lambda] - \frac{1}{2}\log\left|\frac{1}{2}\hat{Q} - 2\Lambda\right| + \frac{n}{2}\log 2\pi,$$

and

$$\Phi(Q) = \mathbb{E}_{u\sim\mathcal{N}(0,Q)}\left[\prod_{a=1}^{n}\Theta(u_a - \kappa)\right].$$

Again, as we did in Theorem 1 proof, we will consider a replica symmetric ansatz.

Let's assume that the maximum of f occurs at replica symmetric values: $Q = (1-q)I + q\mathbf{1}\mathbf{1}^\top$, where $\mathbf{1}$ is a vector of ones, $\hat{Q} = (1-\hat{q})I + \hat{q}\mathbf{1}\mathbf{1}\top$ and we assume that Λ is a diagonal matrix with all equal elements $\Lambda = \lambda I$. Then, $f(Q,\hat{Q},\Lambda)$ can be simplified in terms of $q, \hat{q}, \lambda$ as:

$$f_{RS}(q,\hat{q},\lambda) = \frac{n(n-1)}{4}q\hat{q} + \alpha \log\left(\mathbb{E}_{z\sim\mathcal{N}(0,1)}\left(\mathbb{E}_{u\sim\mathcal{N}(z\sqrt{q},1-q)} f(u)\right)^n\right) - n\lambda$$

$$-\frac{1}{2}\log\left[\left(\frac{1}{2}(1+(n-1)\hat{q}) - 2\lambda\right)\left(\frac{1}{2}(1-\hat{q}) - 2\lambda\right)^{n-1}\right] + \frac{n}{2}\log 2\pi. \tag{26}$$

This can be seen using some ideas from the calculus done in 4.1.6, details are in 4.2.3.

Our goal now is to compute the limit $n \to 0$ of $\frac{1}{n} f_{RS}(q,\hat{q},\lambda)$. We will differentiate with respect to $\hat{q}$ and λ to obtain expressions for $\hat{q}$, λ.

We are going to compute the limit term by term. We start with the basic ones:

$$\frac{1}{n}\frac{n(n-1)}{4}q\hat{q} = \frac{n-1}{4}q\hat{q} \longrightarrow -\frac{1}{4}q\hat{q}.$$

$$-\frac{1}{n}n\lambda \longrightarrow -\lambda$$

$$\frac{1}{n}\frac{n}{2}\log(2\pi) \longrightarrow \frac{1}{2}\log(2\pi)$$

Now we define

$$G(n) = \mathbb{E}_z\left[(p(z;q))^n\right], \qquad p(z;q) = \mathbb{E}_{u\sim\mathcal{N}(z\sqrt{q},\,1-q)}[\Theta(u-\kappa)],$$

to simplify our notation. Taking into account that for a small n we have

$$(p(z;q))^n = e^{\,n\log p(z;q)} \simeq 1 + n\log p(z;q),$$

we can express

$$G(n) = \mathbb{E}_z[1 + n\log p(z;q)] = 1 + n\,\mathbb{E}_z[\log p(z;q)].$$

Now, we use the fact that

$$\log G(n) \simeq n\,\mathbb{E}_z[\log p(z;q)],$$

and taking the limit $n \to 0$ we obtain

$$\frac{1}{n}\alpha\log G(n) \xrightarrow[n\to 0]{} \alpha\,\mathbb{E}_z[\log p(z;q)].$$

We finally compute the limit of the determinant term.

Let us define A_n and B_n as follows:

$$A_n = \frac{1}{2}\big(1 + (n-1)\hat{q}\big) - 2\lambda, \qquad B_n = \frac{1}{2}(1-\hat{q}) - 2\lambda \equiv B.$$

So the term that we want to manipulate is

$$-\frac{1}{2n}\log\!\big(A_n B_n^{n-1}\big) = -\frac{1}{2n}\Big[\log A_n + (n-1)\log B\Big].$$

We write

$$A_n = B + \frac{n}{2}\hat{q},$$

and expand for small n:

$$\log A_n = \log\!\Big(B + \frac{n}{2}\hat{q}\Big) = \log B + \frac{n\hat{q}}{2B} + O(n^2),$$

$$(n-1)\log B = -\log B + n\log B.$$

Therefore,

$$\log A_n + (n-1)\log B = n\left(\log B + \frac{\hat{q}}{2B}\right) + O(n^2),$$

and hence

$$-\frac{1}{2n}\Big[\log A_n + (n-1)\log B\Big] \xrightarrow[n\to 0]{} -\frac{1}{2}\left(\log B + \frac{\hat{q}}{2B}\right).$$

After combining all terms, we have obtained the full replica–symmetric free energy in the limit $n \to 0$:

$$\lim_{n\to 0} \frac{1}{n} f_{RS}(q, \hat{q}, \lambda) = -\frac{1}{4} q\hat{q} + \alpha\, \mathbb{E}_z[\log p(z; q)] - \lambda - \frac{1}{2}\left(\log B + \frac{\hat{q}}{2B}\right) + \frac{1}{2}\log(2\pi),$$

with

$$B = \frac{1}{2}(1 - \hat{q}) - 2\lambda, \qquad p(z; q) = \mathbb{E}_{u\sim\mathcal{N}(z\sqrt{q},\, 1-q)}[\Theta(u - \kappa)].$$

Our next step is to impose the saddle–point conditions by differentiating with respect to $\hat{q}$ and λ and setting the derivatives to zero.

Using $\partial_{\hat{q}} B = -\frac{1}{2}$, we obtain

$$\begin{aligned} \frac{\partial}{\partial \hat{q}}\left[-\frac{1}{4} q\hat{q} - \frac{1}{2}\left(\log B + \frac{\hat{q}}{2B}\right)\right] &= -\frac{q}{4} - \frac{1}{2}\left(\frac{1}{B}\,\partial_{\hat{q}} B\right) - \frac{1}{4}\,\partial_{\hat{q}}\left(\frac{\hat{q}}{B}\right) \\ &= -\frac{q}{4} + \frac{1}{4B} - \frac{1}{4}\left(\frac{1}{B} + \frac{\hat{q}}{2B^2}\right) \\ &= -\frac{q}{4} - \frac{\hat{q}}{8B^2} = \frac{1}{4}\left(-q - \frac{\hat{q}}{2B^2}\right). \end{aligned}$$

The saddle–point condition $\partial f / \partial \hat{q} = 0$ gives

$$-q - \frac{\hat{q}}{2B^2} = 0 \quad \Rightarrow \quad \hat{q} = -2qB^2.$$

Using $\partial_\lambda B = -2$, we find

$$\begin{aligned} \frac{\partial}{\partial \lambda}\left[-\lambda - \frac{1}{2}\left(\log B + \frac{\hat{q}}{2B}\right)\right] &= -1 - \frac{1}{2}\frac{1}{B}\,\partial_\lambda B - \frac{\hat{q}}{4}\,\partial_\lambda\left(\frac{1}{B}\right) \\ &= -1 + \frac{1}{B} - \frac{\hat{q}}{2B^2}. \end{aligned}$$

Imposing the saddle–point condition $\partial f / \partial \lambda = 0$ yields

$$-1 + \frac{1}{B} - \frac{\hat{q}}{2B^2} = 0.$$

Next step is to solve. Substituting $\hat{q} = -2qB^2$ into the second saddle–point equation gives

$$-1 + \frac{1}{B} + q = 0 \quad \Rightarrow \quad B = \frac{1}{1-q}.$$

Hence

$$\hat{q}(q) = -\frac{2q}{(1-q)^2}.$$

Using the definition of B,

$$B = \frac{1}{2}(1 - \hat{q}) - 2\lambda,$$

we solve for λ:

$$\lambda = \frac{1}{4}(1-\hat{q}) - \frac{B}{2} = \frac{q^2+2q-1}{4(1-q)^2}.$$

We proceed by making the potential depend only on q. Substituting $\hat{q}(q)$, $\lambda(q)$ and $B(q) = \frac{1}{1-q}$ back into the free energy and simplifying, we obtain

$$\lim_{n\to 0} \frac{1}{n} f_{RS}(q, \hat{q}(q), \lambda(q)) = \alpha\, \mathbb{E}_z[\log p(z;q)] + \frac{1+q}{4(1-q)} + \frac{1}{2}\log(1-q) + \text{const.}$$

Up to an additive constant, we can, therefore, write the effective potential as

$$\Phi(q) = \alpha\, \mathbb{E}_z[\log p(z;q)] + \frac{1+q}{4(1-q)} + \frac{1}{2}\log(1-q)$$

which now depends only on the physical order parameter q.

Next, let us simplify the expression by eliminating irrelevant additive constants. Using the identity

$$\frac{1+q}{4(1-q)} = \frac{1}{2}\frac{1}{1-q} - \frac{1}{4},$$

we see that the term $-\frac{1}{4}$ does not depend on q and can therefore be discarded. Moreover, since

$$p(z;q) = \mathbb{P}\left(u \geq \frac{\kappa - \sqrt{q}\, z}{\sqrt{1-q}}\right), \qquad u \sim \mathcal{N}(0,1),$$

the effective potential can be written, up to an additive constant, as

$$\Phi(q) = \alpha\, \mathbb{E}_z\left[\log \mathbb{P}\left(u \geq \frac{\kappa - \sqrt{q}\, z}{\sqrt{1-q}}\right)\right] + \frac{1}{2}\frac{1}{1-q} + \frac{1}{2}\log(1-q),$$

which is exactly what we wanted to prove.

□

4.2.3 Symmetry ansatz for perceptron

We first simplify the term $\frac{1}{2}\sum_{a<b} Q_{ab}\hat{Q}_{ab}$. There are $\frac{n(n-1)}{2}$ pairs $a<b$. For each one of these, $Q_{ab} = q$ and $\hat{Q}_{ab} = \hat{q}$. Therefore,

$$\frac{1}{2}\sum_{a<b} Q_{ab}\hat{Q}_{ab} = \frac{1}{2}\cdot\frac{n(n-1)}{2}\cdot q\hat{q} = \frac{n(n-1)}{4}\hat{q}q.$$

We define the matrix $A = \frac{1}{2}\hat{Q} - 2\Lambda$ and we calculate it's determinant. $\hat{Q}$ has eigenvalues $1+(n-1)\hat{q}$ with multiplicity 1 and $1-\hat{q}$ with multiplicity $n-1$. Therefore, matrix A has eigenvalues $\frac{1}{2}(1+(n-1)\hat{q}) - 2\lambda$ with multiplicity 1 and $\frac{1}{2}(1-\hat{q}) - 2\lambda$ with multiplicity $n-1$. The determinant of M is the product of the eigenvalues. So,

$$\left|\frac{1}{2}\hat{Q} - 2\Lambda\right| = \left(\frac{1}{2}(1 + (n-1)\hat{q}) - 2\lambda\right)\left(\frac{1}{2}(1 - \hat{q}) - 2\lambda\right)^{n-1}.$$

For the term $\alpha \log \Phi(Q)$, we will use the identity:

$$\mathbb{E}_{u \sim \mathcal{N}(0,Q)}\left[\prod_{a=1}^{n} f(u_a)\right] = \mathbb{E}_{z \sim \mathcal{N}(0,1)}\left(\mathbb{E}_{u \sim \mathcal{N}(z\sqrt{q}, 1-q)} f(u)\right)^{n}.$$

In our case, $f(u_a) = \Theta(u_a - \kappa)$.

Finally, $tr[\Lambda] = n\lambda$. Putting it all together, we obtain the expected expression.

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